

Notes on Classical Electron Dynamics: Lorentz–Dirac Equation

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ABSTRACT

These notes reorganize, twenty years later, the classical-electron part of an undergraduate thesis written in 2006, and set it against the literature it was drawing on. The material is arranged as one line of argument. The runaway appears with Lorentz, survives Dirac’s much better derivation unchanged, and is then attacked along the two branches Eliezer identified in 1947: change the coefficient of the radiation-reaction term, or raise the order of the equation. Each branch is followed on its own terms and scored against criteria fixed before either is on the table. Two things are recorded as they arise — where a result the 2006 text reached numerically had already been obtained analytically, and where a step it relies on is in tension with what is now known. No new results are claimed, and the twenty-years-on assessment is left for a later section.

1. THE LORENTZ EQUATION AND THE RUNAWAY PROBLEM

Conventions, fixed here and used throughout. Electrodynamic quantities are in Gaussian units. The factor c is kept explicit in formulae rather than set to one, and where it is convenient to drop it — in the tube integral of Sec. 2.1.2 and in the non-relativistic reductions of Sec. 5 — that is said locally and c is restored at the end. The metric signature is $(-, +, +, +)$, so that $u^\alpha u_\alpha = -1$. The world line is $z^\alpha(\tau)$ with τ the proper time, and $u^\alpha = dz^\alpha/d\tau$, $a^\alpha = du^\alpha/d\tau$; an overdot means $d/d\tau$ except in explicitly non-relativistic passages, where it means d/dt . Differentiating $u \cdot u = -1$ gives $u \cdot a = 0$ and then $u \cdot \dot{a} = -a \cdot a$, both of which are used constantly and neither of which is flagged again.

One number is worth fixing at the outset, because every section below uses it. The characteristic time of the electron is $\tau_0 = 2e^2/3mc^3$, which in Gaussian units is 6.27×10^{-24} s. [D. Zhang \(2006\)](#) declares Gaussian units but evaluates this quantity without the factor $1/4\pi\epsilon_0$, obtaining 10^{-33} s instead. The corrected value is used here from the start, and nothing else is made of the discrepancy.

1.1. *Electrodynamics does not supply an equation of motion for the charge*

Classical electrodynamics rests on three pillars: the conservation laws, chief among them the conservation of charge, Maxwell’s equations, and the Lorentz force law ([Z. Zhang 1979](#)). Maxwell’s equations fix the field produced by a given distribution of charge and current. The Lorentz force law fixes the force exerted on a charge by a given field. Between them the two appear to account for everything an electrodynamical problem could ask for, and they do not: neither supplies the equation of motion of the charge itself.

The gap is structural rather than an oversight. To follow the motion of a charge one needs the force acting on it, hence the total field at its location, and the total field includes the field that the charge itself produces. Maxwell’s equations say how that self-field is generated but not how it acts back. The Lorentz force law says how a field acts on a charge but is silent on whether the charge’s own

field belongs among the fields that act. Electrodynamics is complete as an account of how charges make fields and of how fields push charges, and it is incomplete exactly where the two meet on the same particle. Closing that gap is the subject of everything below.

1.2. *Three historical routes*

Any attempt to close the gap must begin by deciding what a charge is, and two pictures divide the subject between them. In the first the charge sits at a point. Its own field then diverges as the point is approached, and with the field the Lorentz force, so the force expression loses its meaning exactly where it is needed. One may object that the Lorentz force was only ever intended for external fields and that the self-field is simply not subject to it. But the objection buys nothing, because it leaves the self-force undetermined and the equation of motion no closer than before. That is the standing difficulty of the point picture. In the second the charge is spread over a finite region and treated as a continuous medium. Every quantity is then finite and the self-force is a well-defined integral over the distribution. The price is that like charges repel, so a body carrying charge of one sign must fly apart. Bound charges nevertheless exist, and electrodynamics does not say what holds them together.

Three attempts, spread over thirty years, took these pictures in turn. [H. Poincaré \(1906\)](#) worked in the extended picture and supplied the missing agent by hand, adding to the electromagnetic stress–energy tensor a binding tensor called the Poincaré stresses, chosen so that the total is in equilibrium and the charged body holds. The construction does what is asked of it, but it postulates a force for the sole purpose of removing a single contradiction, with no other consequence and no independent evidence for its existence. In the century since, few have followed it.

[H. A. Lorentz \(1909\)](#) took a position between the two pictures: the electron is a sphere of finite radius carrying a uniform charge. This is concrete enough to yield a definite equation of motion, which is the subject of [Sec. 1.3](#).

[P. A. M. Dirac \(1938\)](#) returned to the point picture, but gave up computing the self-force as the mutual action of one part of the charge on another. He imposed momentum–energy conservation on a thin tube surrounding the world line and read the equation of motion off the balance, so that the divergent self-field enters only through quantities that cancel. This is the route the rest of these notes follow. It is set out in [Sec. 2](#).

These notes adopt the point picture, for the reasons that recommended it to Dirac. It is technically uniform. The model carries no internal structure that has to be specified in advance. The equation of motion comes out in a fixed form. The construction also generalizes, both to curved backgrounds and to the corresponding gravitational problem. The cost is that infinite quantities appear at intermediate stages, so the force has to be extracted indirectly. This can be done either by balancing finite quantities, as in the conservation method, or by separating off a singular part and discarding it. In what sense those operations are legitimate is a question that runs through everything below.

1.3. *The Lorentz model and its three defects*

Lorentz’s electron is a sphere of finite radius carrying a uniform charge. It is far enough into the extended picture that nothing diverges, and structureless enough to remain tractable. What holds the charge together is left unanswered, which is precisely the question the Poincaré stresses were invented to settle. In motion the sphere is contracted into an ellipsoid.

Two forces then act on it. One is external. The other is a self-force, produced by the action of the charge elements on one another. At rest this self-force vanishes by symmetry. In motion it does

not, and the reason is retardation: the field with which one element acts on a second is the field it carried at an earlier time, and the reaction of the second on the first is delayed in the same way, so the two no longer cancel. The charge elements taken by themselves therefore violate Newton's third law. Nothing is genuinely lost. The missing momentum resides in the field. But the failure is a fair measure of how far the problem has already moved away from mechanics.

1.3.1. Deriving the self-force directly

The calculation itself is not deep. The work is all in the expansion, and it is standard (see, e.g., [J. D. Jackson 1999](#); [F. Rohrlich 2007](#)). Let the charge be distributed with density ρ over a rigid body of size R , moving slowly enough that magnetic contributions may be dropped to the order kept. The self-force is the double integral of the retarded interaction over all pairs of charge elements,

$$\mathbf{F}_{\text{self}} = \iint d^3r d^3r' \rho(\mathbf{r}) \rho(\mathbf{r}') \mathbf{K}(\mathbf{r}, \mathbf{r}'; t - s), \quad s = \frac{|\mathbf{r} - \mathbf{r}'|}{c}, \quad (1)$$

in which the kernel $\mathbf{K}$ is evaluated not at the present time but one light-crossing time s earlier. Since $s \sim R/c$ is small, expand every retarded quantity about the present instant,

$$f(t - s) = f(t) - s \dot{f}(t) + \frac{s^2}{2} \ddot{f}(t) - \frac{s^3}{6} \dddot{f}(t) + \dots, \quad (2)$$

and collect. Each additional power of s costs one factor of R/c and buys one more time derivative of $\mathbf{v}$. The series therefore organizes itself by derivative order. The term containing $d^n \mathbf{v}/dt^n$ carries R^{n-2} . The zeroth term is the static self-repulsion, which vanishes by symmetry for a rigid symmetric body, and what survives is

$$\mathbf{F}_{\text{self}} = -\frac{4U_0}{3c^2} \dot{\mathbf{v}} + \frac{2e^2}{3c^3} \ddot{\mathbf{v}} + \sum_{n \geq 3} \alpha_n \frac{e^2 R^{n-2}}{c^{n+1}} \frac{d^n \mathbf{v}}{dt^n}, \quad (3)$$

where U_0 is the electrostatic self-energy and the α_n are pure numbers fixed by the shape of the distribution.

The three groups of terms meet three quite different fates, set out in [Table 1](#), and the whole of the subject descends from that split.

Term	Coefficient scales as	What becomes of it
$n = 1, \propto \dot{\mathbf{v}}$	R^{-1} , divergent	absorbed into the mass
$n = 2, \propto \ddot{\mathbf{v}}$	R^0 , model-independent	survives as the radiation reaction
$n \geq 3, \propto d^n \mathbf{v}/dt^n$	R^{n-2} , vanishing	lost in the point limit

Table 1. The three groups of terms in the expansion [Eq. \(3\)](#), and what becomes of each as the radius of the body is taken to zero. Only the middle row survives, and the classical theory of the electron is the theory of that one row.

The first row is the divergence, and it is harmless only because of its form. Being proportional to $\dot{\mathbf{v}}$, it cannot be distinguished from mass times acceleration. It can therefore be hidden in the mass,

$$m = m_{\text{bare}} + \frac{4U_0}{3c^2}. \quad (4)$$

This is the first appearance in the subject of what would later be called mass renormalization, and the source of the first defect below.

The second row is the one that matters, and its coefficient cannot depend on how the charge is arranged. With no power of R available there is nothing left to build a coefficient from except e and c , so the combination is fixed up to a pure number. A sphere, a shell and a dumbbell all return $2e^2/3c^3$, while each returns its own coefficient for the divergent term. That asymmetry is the reason the radiation-reaction term is believed and the mass term is renormalized away.

The third row is the casualty. Those terms are the only place where the internal structure of the electron would ever have shown itself, and the point limit destroys precisely that information.

Substituting Eq. (3) into $m_{\text{bare}}\dot{\mathbf{v}} = \mathbf{F} + \mathbf{F}_{\text{self}}$ and using Eq. (4) gives

$$\boxed{m\dot{\mathbf{v}} - \frac{2e^2}{3c^3}\ddot{\mathbf{v}} + \dots = \mathbf{F}} \quad (\text{AL})$$

now usually called the Abraham–Lorentz equation (M. Abraham 1905; H. A. Lorentz 1909), in which m is the observed mass and everything from the second term on the left is the self-force.

1.3.2. The same result from energy balance

There is a much shorter route, which avoids the charge distribution altogether and therefore never meets the divergent mass. Take the Larmor formula (J. Larmor 1897) for the radiated power, $P = (2e^2/3c^3)\dot{\mathbf{v}} \cdot \dot{\mathbf{v}}$, and demand that the work done by the self-force account for the energy radiated over an interval,

$$\int_{t_1}^{t_2} \mathbf{F}_{\text{self}} \cdot \mathbf{v} dt = - \int_{t_1}^{t_2} \frac{2e^2}{3c^3} \dot{\mathbf{v}} \cdot \dot{\mathbf{v}} dt. \quad (5)$$

Integrating the right-hand side by parts,

$$- \int_{t_1}^{t_2} \frac{2e^2}{3c^3} \dot{\mathbf{v}} \cdot \dot{\mathbf{v}} dt = - \frac{2e^2}{3c^3} [\dot{\mathbf{v}} \cdot \mathbf{v}]_{t_1}^{t_2} + \int_{t_1}^{t_2} \frac{2e^2}{3c^3} \ddot{\mathbf{v}} \cdot \mathbf{v} dt. \quad (6)$$

If the boundary term vanishes, as it does for periodic motion or whenever $\dot{\mathbf{v}} \cdot \mathbf{v}$ takes the same value at both ends, the two integrands may be compared:

$$\mathbf{F}_{\text{self}} = \frac{2e^2}{3c^3} \ddot{\mathbf{v}}, \quad (7)$$

which is the self-force of Eq. (AL) in three lines. The mass never enters, so it stays where classical mechanics put it.

Two things should be said about what this shortcut costs. First, the conclusion is drawn from equality of integrals over a restricted class of intervals, not from any pointwise identity, so it establishes an average effect and not an instantaneous law. The third and higher derivatives of $\mathbf{v}$, which the direct calculation delivers in Eq. (3), are simply invisible to it. Second, and more interestingly, the term thrown away is not junk. The boundary term $(2e^2/3c^3)\dot{\mathbf{v}} \cdot \mathbf{v}$ is the Schott energy (G. A. Schott 1912), and the piece of the covariant reaction force whose time derivative it is will turn out to be the Schott term. Everything awkward about energy balance in Sec. 3 is already present here. The radiated power and the work done by the reaction force refuse to match instant by instant. Uniformly accelerated motion radiates while feeling no reaction at all. Both puzzles sit in the term this derivation is allowed to discard because the motion was assumed periodic.

1.3.3. Three defects

Three objections can be raised against Eq. (AL).

The mass is entirely electromagnetic, and comes out with the wrong coefficient. The absorbed term gives $m = 4U_0/3c^2$, where U_0 is the electrostatic self-energy, finite here because the radius is. The factor $4/3$ is itself a symptom: a covariant accounting of the momentum of a bound charge distribution yields U_0/c^2 , and the extra third appears only because the electromagnetic stress has been counted while whatever holds the sphere together has not. It is the same omission as before, now showing up as a number. E. Fermi (1922) and W. Wilson (1936) later showed that the coefficient becomes unity once the binding stresses are included. The deeper objection survives the repair. If inertia is electromagnetic in origin, a neutral particle should have no mass, and neutral particles have mass. Inertial mass cannot be a manifestation of charge. At most one may say that charge brings mass with it. Nor does Lorentz's derivation claim otherwise on its own terms, since the U_0 appearing in it is electromagnetic energy alone and contains no mechanical rest energy. The energy-balance derivation of Sec. 1.3.2 sidesteps the whole question, which is its chief attraction.

The equation is not covariant. Equation (AL) is written with d/dt in a particular frame. The repair is due to L. D. Landau & E. M. Lifshitz (1975) and is disarmingly short. Replace d/dt by $d/d\tau$ and guess $m \dot{u}^\alpha = F^\alpha + k \ddot{u}^\alpha$. The guess fails at once, because contracting with u_α and using $u^\alpha u_\alpha = -1$, whose first two derivatives give $u \cdot a = 0$ and $u \cdot \dot{a} = -a \cdot a$, leaves $-k(a \cdot a)$ on the right where zero is required. The cure is to add a term along u^α and let the constraint fix its coefficient: demanding $u_\alpha(\dot{a}^\alpha + C u^\alpha a \cdot a) = 0$ gives $C = -1$, so that the reaction force must be

$$\frac{2e^2}{3c^3} (\dot{a}^\alpha - u^\alpha a^\beta a_\beta) = \frac{2e^2}{3c^3} (\delta^\alpha_\beta + u^\alpha u_\beta) \dot{a}^\beta. \quad (8)$$

This is exactly the term that Dirac's far longer construction produces in Sec. 2. Covariance alone, applied to a nonrelativistic result, very nearly hands over the answer. This is worth keeping in mind when judging how much the tube derivation really establishes.

A free electron accelerates itself without bound. This is the serious defect, and it is not repaired anywhere in these notes. It is the subject of Sec. 1.4.

1.4. The Abraham-Lorentz equation and its runaway solutions

Set $\mathbf{F} = 0$ in Eq. (AL), keep only the two terms written, and reduce to one dimension. The coefficient of the self-force term is m times a time,

$$\tau_0 = \frac{2e^2}{3mc^3} = 6.27 \times 10^{-24} \text{ s} \quad (\text{electron}), \quad (9)$$

which will reappear in every section below, and the equation of motion of a free electron collapses to a first-order equation for the acceleration alone,

$$\dot{a} = \frac{a}{\tau_0}, \quad \text{whence} \quad a = C e^{t/\tau_0}. \quad (10)$$

Both branches of this solution are unsatisfactory, and for different reasons. If $C \neq 0$, an electron with nothing whatever acting on it accelerates itself, and does so on the timescale τ_0 , which is to say it gains a factor of e in acceleration every $\sim 10^{-23}$ seconds. In the nonrelativistic equation the

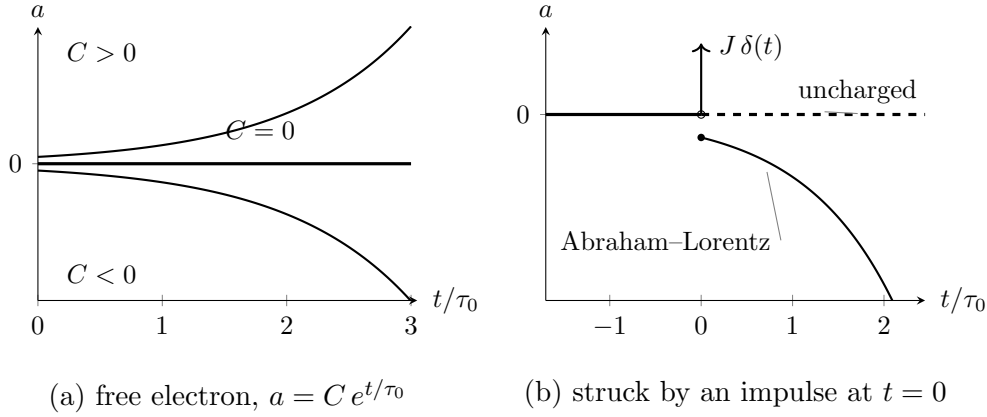


Figure 1. The two faces of the third-order term. (a) A free electron obeying Eq. (AL) has $a = C e^{t/\tau_0}$. Every $C \neq 0$ runs away; the physical solution is the single value $C = 0$, and nothing in the initial-value problem selects it, since the equation is first order in a and admits any $a(0)$. (b) An impulse delivered at $t = 0$. An uncharged particle takes the blow, jumps in velocity, and returns to zero acceleration: the force is over and so is the acceleration. The electron does the opposite. Its acceleration jumps to $\kappa = -J/m\tau_0$ and grows without bound long after nothing is acting on it, and, since κ carries the sign opposite to J , it runs away against the blow that started it. The velocity, by contrast, is continuous at $t = 0$ here: it is the acceleration that jumps.

velocity grows without bound. In the covariant version of Sec. 3 it approaches c instead, but the character of the solution is unchanged. No intuition about free particles survives this, and no agent is available to be held responsible for the energy.

If $C = 0$ then $a = 0$, the electron moves uniformly, and Newton's first law is recovered. One may say that the Abraham–Lorentz equation tells us that a free electron moves exactly as an uncharged body does. The physical content is unobjectionable. What is objectionable is the standing of the condition. Equation (10) is first order in a and therefore admits any initial value $a(0)$. The physical solution is picked out by setting $a(0) = 0$, and the only reason offered for doing so is that every other choice misbehaves as $t \rightarrow \infty$. A condition imposed at infinity to restrict data at $t = 0$ has no warrant in the initial-value problem it is being applied to. This remains true no matter how reasonable the surviving solution looks.

The impulse problem sharpens the objection into something concrete. Write the electron's equation with an impulsive force of strength J delivered at $t = 0$,

$$ma - m\tau_0 \dot{a} = J \delta(t), \quad (11)$$

and compare with the corresponding classical problem. A free mass point struck by an impulse jumps discontinuously in velocity and has zero acceleration immediately afterwards: the blow is over, and so is the acceleration. Solving Eq. (11) with the electron moving uniformly beforehand gives instead

$$a = \begin{cases} 0, & t < 0, \\ \kappa e^{t/\tau_0}, & t > 0, \end{cases} \quad \kappa = -\frac{J}{m\tau_0}. \quad (12)$$

The electron does not stop accelerating when the blow ends. It accelerates ever harder, without limit, long after nothing is acting on it. Since κ carries the opposite sign to J , it runs away in the direction

opposite to the blow that started it. This is the runaway, or acceleration collapse, in its starkest form. No choice of initial data avoids it here, because the requirement of uniform motion before the pulse has already been spent.

The root of the behaviour is visible in the order of the equation. Newton's equation of motion contains the acceleration and nothing beyond it, so when the force stops the acceleration stops with it. Equation (AL) contains derivatives of the acceleration, so the state of the electron at a given instant includes how fast its acceleration is changing, and that quantity survives the force that produced it. What sustains it, physically, is the electron's own electromagnetic field. The defect is therefore not an artefact of the sphere model or of the expansion used to reach Eq. (AL). Any equation of motion that accounts for the self-field by adding a derivative of the acceleration will inherit it. Dirac's equation, derived on entirely different principles in Sec. 2, inherits it exactly.

1.5. *Criteria for a physically acceptable equation of motion*

Everything from here on consists of proposals to replace Eq. (AL) with something better behaved, and proposals cannot be judged without a standard to judge them by. It is worth fixing that standard now, before any candidate is on the table, so that it cannot be quietly adjusted to fit whichever equation is under discussion. Three requirements are imposed throughout these notes.

1. **The motion must follow from the equation in the ordinary way.** The particle should obey a differential equation that can be solved by the usual procedure. One supplies initial or boundary data, integrates, and obtains the state at all later times. The data supplied must themselves be data the physical situation could actually provide. This is a weaker demand than it may appear, and Sec. 1.4 has already shown Eq. (AL) failing it: the equation is solvable, but the solution that survives is picked out by a condition imposed at $t \rightarrow \infty$, and no experimenter preparing an electron at $t = 0$ has access to that condition.
2. **A stable state must remain stable under a small disturbance.** The concrete test used throughout is the one already applied in Sec. 1.4: take a particle in uniform motion, strike it with a pulse, and ask what it does afterwards. An acceptable equation returns it to uniform motion. It is also acceptable for the particle to settle into a regular variation confined to a bounded range of speeds. This weaker alternative is included deliberately, because Sec. 5 produces solutions of exactly this kind. What is not acceptable is for the particle to accelerate without limit toward c , or to swing back and forth between $+c$ and $-c$. Both are read here as signs that the equation has no stable state to return to, rather than as physics.
3. **The ordinary physical laws must hold.** Energy conservation and causality are not negotiable, and an equation that purchases good behaviour by breaking either has not solved the problem but relocated it. This criterion does the least work in the pages that follow and is the easiest to forget, which is why it is worth writing down. The sign of a radiation-reaction term, for instance, is exactly the kind of thing one may change to remove a runaway without noticing what it does to the energy budget.

These three are the scorecard. Sections 3, 4 and 5 each close by returning to them. Two features of the list are worth noticing before it is used. It is stated entirely in terms of the behaviour of solutions and says nothing about the form an equation should take, which is deliberate, since the strategy of the sections that follow is precisely to alter the form and inspect what comes out. And it asks that

a charged particle behave, in these respects, like an ordinary mechanical one. That expectation is reasonable and widely shared, but it is worth flagging as an assumption rather than a deduction.

2. THE LORENTZ–DIRAC EQUATION: DERIVATION

A word on the name before the derivation. Throughout these notes *Lorentz–Dirac*, abbreviated LD and also found in the literature as Abraham–Lorentz–Dirac, means the classical equation of motion of a point charge derived below and tagged (LD). It never means Dirac’s relativistic wave equation, with which it has nothing in common but its author.

2.1. The world-tube momentum–energy balance

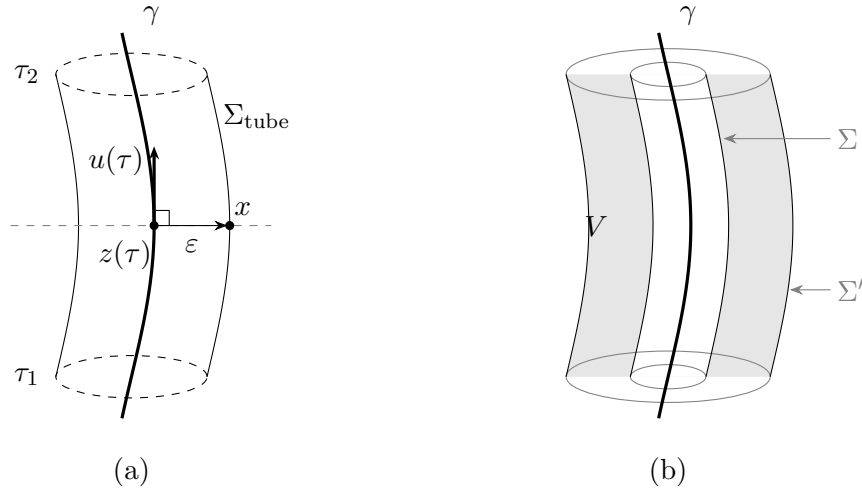


Figure 2. Dirac’s construction. (a) The tube. At each proper time the point $z(\tau)$ on the world line γ carries a sphere of radius ε , drawn in the hyperplane of simultaneity of $z(\tau)$ — the dashed line — so that the separation $x - z(\tau)$ to a point x of the tube is orthogonal to the four-velocity $u(\tau)$, as in Eq. (15). Dragging that sphere along γ from τ_1 to τ_2 sweeps out Σ_{tube} . The end caps are drawn dashed as a reminder that the integration is carried out over the wall alone; see Sec. 2.5. (b) Why the choice of tube does not matter. Two tubes Σ and Σ' enclose the same world line, and the region V between them contains no charge, so the flux of $T^{\mu\nu}$ through the two is the same by Eq. (14). The balance may therefore be read on any surface, including one that keeps well away from the singular world line.

Dirac’s advance was a change of question. Lorentz’s route, followed in Sec. 1.3, asks what force the charge exerts on itself and answers by summing the retarded action of one charge element on another. That sum has no meaning for a point charge, since the self-field diverges on the world line. Dirac keeps the point charge and declines to ask about the force at all. He asks instead where the energy and momentum go. Whatever the particle radiates must be paid for out of the particle’s own energy and momentum. That balance can be struck without ever evaluating a field on the world line. The divergent self-field still enters the calculation, but only through differences, and the differences are finite.

2.1.1. The tube, and why its shape is immaterial

Surround the world line γ by a three-dimensional tube Σ . The four-momentum crossing it is

$$\Delta P^\mu = \int_{\Sigma} T^{\mu\nu} d\Sigma_\nu. \quad (13)$$

Let Σ and Σ' both enclose γ , and let V be the region between them. That region contains no charge, so the stress–energy tensor is conserved throughout it, and

$$\Delta P'^\mu - \Delta P^\mu = \oint_{\partial V} T^{\mu\nu} d\Sigma_\nu = \int_V T^{\mu\nu}{}_{,\nu} dV = 0. \quad (14)$$

This is the whole reason the strategy beats Lorentz’s. It deserves to be stated on its own. A force must be evaluated somewhere, and for a point charge every candidate location is a place where the field is infinite. A balance need not be evaluated anywhere in particular. It may be read off on any surface at all, including surfaces that keep a comfortable distance from the singularity. Equation (14) guarantees that the reading is the same. See Fig. 2(b).

Dirac’s tube is the one adapted to the instantaneous rest frame. For a point x on the tube, let $z(\tau)$ be the point of γ satisfying

$$[x^\mu - z^\mu(\tau)] u_\mu(\tau) = 0, \quad [x - z(\tau)] \cdot [x - z(\tau)] = \varepsilon^2, \quad (15)$$

the separation being spacelike. The first condition places x on the hyperplane of simultaneity of $z(\tau)$, so that “at the same moment” means at the same moment for the particle. The second fixes the radius. Together they describe a sphere of radius ε drawn in the instantaneous rest frame and dragged along the world line, sweeping out $\Sigma_{\text{tube}} = \{x(\tau, \varepsilon, \Omega); \tau_1 \leq \tau \leq \tau_2\}$, as in Fig. 2(a).

2.1.2. The surface element

Set $c = 1$ for the rest of this subsection and work in the instantaneous rest frame at proper time τ , restoring c at the end. Let n^μ be the unit spacelike vector pointing from $z(\tau)$ to the point x of the tube, so that $x = z(\tau) + \varepsilon n$ with $n \cdot u = 0$ and $n \cdot n = 1$. Three angular integrals are all that will be needed:

$$\oint d\Omega = 4\pi, \quad \oint n^i d\Omega = 0, \quad \oint n^i n^j d\Omega = \frac{4\pi}{3} \delta^{ij}. \quad (16)$$

Differentiate $x = z + \varepsilon n$ along the world line at fixed Ω . The vector n is Fermi–Walker transported, so $\dot{n}^\mu = (a \cdot n) u^\mu$. Thus

$$\frac{\partial x^\mu}{\partial \tau} = u^\mu (1 + \varepsilon a \cdot n). \quad (17)$$

Each of the two angular directions contributes a factor ε . Thus

$$d\Sigma_\nu = n_\nu \varepsilon^2 (1 + \varepsilon a \cdot n) d\Omega d\tau. \quad (18)$$

The factor $1 + \varepsilon a \cdot n$ is the one thing in this subsection worth memorizing. It is there because the tube is dragged along a world line that is accelerating. Successive cross-sections are not parallel, and the wall stretches on the side the particle is accelerating towards. It is also easy to write the surface element without it, and everything that distinguishes this calculation from the charged sphere of Sec. 1.3 is contained in it.

2.1.3. The field near the world line

Expanded about $z(\tau)$ in the rest frame, the retarded field is

$$\mathbf{E} = \frac{e}{\varepsilon^2} \mathbf{n} - \frac{e}{2\varepsilon} [\mathbf{a} + (\mathbf{a} \cdot \mathbf{n})\mathbf{n}] + \frac{2e}{3} \dot{\mathbf{a}} + O(\varepsilon), \quad (19)$$

while $\mathbf{B}$, which vanishes for a charge at rest, begins only at $O(\varepsilon^{-1})$. The surface element carries ε^2 , so a product of two fields must reach $O(\varepsilon^{-3})$ to leave anything divergent behind. Since $\mathbf{B}\mathbf{B}$ is only $O(\varepsilon^{-2})$, the magnetic field cannot contribute to the divergent term at all, and the whole of the next subsection is an exercise with $\mathbf{E}$. The four-momentum leaving the tube is

$$\frac{dP^\mu}{d\tau} = - \oint T^{\mu\nu} n_\nu \varepsilon^2 (1 + \varepsilon a \cdot n) d\Omega, \quad T^{\mu\nu} = \frac{1}{4\pi} [F^{\mu\alpha} F^\nu{}_\alpha - \tfrac{1}{4} \eta^{\mu\nu} F^{\alpha\beta} F_{\alpha\beta}]. \quad (20)$$

2.1.4. The divergent term, and what becomes of the 4/3

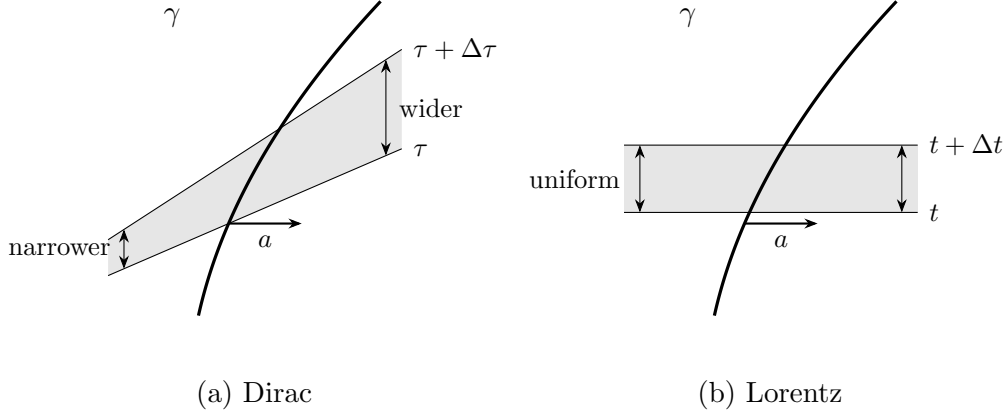


Figure 3. Where the 4/3 goes. Both panels show the same accelerating world line γ and the same slab of it, and differ only in what is being called “at the same moment”. (a) Dirac’s tube is built from the particle’s own succession of rest frames. Because the velocity changes along γ , successive slices are not parallel: they fan open on the side the particle is accelerating towards, and the slab is wider there. That fanning is the factor $1 + \varepsilon a \cdot n$ of Eq. (18). (b) Collecting field momentum at constant time in one chosen frame gives parallel slices and a slab of uniform thickness, with no such factor. The wedge by which the two shaded regions differ is worth exactly one third of the electromagnetic mass, which is the whole of the discrepancy between Eqs. (4) and (24).

Two products of Eq. (19) reach the required order. Take the Coulomb term against itself first.

$$T^{ij} n_j \Big|_{\text{CC}} = \frac{1}{4\pi} \frac{e^2}{\varepsilon^4} [n^i n^j - \tfrac{1}{2} \delta^{ij}] n_j = \frac{e^2}{8\pi \varepsilon^4} n^i. \quad (21)$$

At order ε^{-2} this integrates to nothing, since $\oint n^i d\Omega = 0$. An isotropic Coulomb field pushes equally in every direction. Against the tilt factor it survives:

$$- \oint \frac{e^2}{8\pi \varepsilon^4} n^i \varepsilon^3 (a \cdot n) d\Omega = - \frac{e^2}{8\pi \varepsilon} \cdot \frac{4\pi}{3} a^i = - \frac{e^2}{6\varepsilon} a^i. \quad (22)$$

The second product is the Coulomb term against the acceleration field, which is the one a sphere would have too:

$$- \oint \frac{1}{4\pi} \left[- \frac{e^2}{2\varepsilon^3} a^i - \frac{e^2}{2\varepsilon^3} (a \cdot n) n^i \right] \varepsilon^2 d\Omega = \frac{e^2}{2\varepsilon} \left(1 + \tfrac{1}{3} \right) a^i = \frac{2e^2}{3\varepsilon} a^i. \quad (23)$$

Writing $U_0 = e^2/2\varepsilon$ for the electrostatic energy of a shell of radius ε , the coefficient here is $4U_0/3$: the same 4/3 the sphere produced in Sec. 1.3. It is arrived at in the same way, by ignoring the tilt.

Adding Eqs. (22) and (23),

$$\left. \frac{dP^i}{d\tau} \right|_{\text{div}} = \left(\frac{2}{3} - \frac{1}{6} \right) \frac{e^2}{\varepsilon} a^i = \frac{e^2}{2\varepsilon} a^i = U_0 a^i. \quad (24)$$

The coefficient is 1, not 4/3, and the two derivations in these notes therefore disagree about the electromagnetic mass: Eq. (4) absorbed $4U_0/3c^2$, while Eq. (24) absorbs U_0/c^2 . Neither contains an error, and it is worth being clear about what the disagreement is.

Section 1.3 recorded the 4/3 as the mark of having counted the electromagnetic stress while omitting whatever holds the sphere together. Dirac's calculation has no sphere to hold together and no binding stress to omit, and it returns U_0/c^2 . What produces the difference is the tilt factor of Eq. (18), and the tilt factor is a statement about simultaneity. Lorentz gathers the field momentum on a hyperplane of constant time in one chosen frame. Dirac gathers it on a surface assembled from the particle's own succession of rest frames. The gap between those two ways of deciding what counts as "at the same moment" is precisely the missing third.

The 4/3 was never a statement about electrons. It was a statement about a choice of simultaneity, and it disappears as soon as the choice is made covariantly.

2.1.5. The finite term, and two checks

The cross of the Coulomb field with the radiative term $\frac{2e}{3}\dot{\mathbf{a}}$ gives, after the first and third pieces of $T^{ij}n_j$ cancel against each other,

$$-\oint \frac{1}{4\pi} \frac{2e^2}{3\varepsilon^2} \dot{a}^i \varepsilon^2 d\Omega = -\frac{2e^2}{3} \dot{a}^i. \quad (25)$$

The finite pieces not written out are the acceleration field against itself, $\mathbf{B}$ against itself, and the tilt correction to Eq. (23). They contain no $\dot{a}$, and assemble into the term along u^μ that $u \cdot dP/d\tau = 0$ requires. Restoring c ,

$$\frac{dP^\mu}{d\tau} = \frac{e^2}{2\varepsilon c^2} a^\mu - \frac{2e^2}{3c^3} (\dot{a}^\mu - u^\mu a^\beta a_\beta), \quad (26)$$

P^μ being the four-momentum carried out of the tube.

Two things can be checked without redoing anything. Contracting Eq. (26) with u_μ , and using $u \cdot u = -1$ and $u \cdot \dot{a} = -a \cdot a$, gives zero, as it must. And the component along u^μ is $(2e^2/3c^3) a \cdot a$, the Larmor power. Thus the radiated energy is present with the coefficient that Sec. 1.3.2 obtained from energy balance, by an argument that never mentioned a tube.

What Eq. (26) is not is an equation of motion. It records how much four-momentum the *field* carries out of the tube. Turning that into a statement about the particle requires saying what the particle's own four-momentum is, and nothing in the calculation supplies it. That freedom is the subject of Sec. 2.2.

2.2. The undetermined four-vector B^μ

Total four-momentum is conserved, so the particle's own four-momentum changes at the rate at which the external field supplies it, less the rate at which the field carries it out of the tube,

$$\frac{dP_{\text{mech}}^\mu}{d\tau} = F_{\text{ext}}^\mu - \frac{dP^\mu}{d\tau}, \quad (27)$$

with $dP^\mu/d\tau$ given by Eq. (26). Everything on the right is known. The left is not, because nothing computed so far says what P_{mech}^μ is.

This is the bill for the trick of Sec. 2.1.1. The tube integral was arranged to be insensitive to what lies inside it. That insensitivity is exactly why the divergent self-field never had to be evaluated. But the particle lies inside it. The construction is silent about the interior by design, and it is silent about the particle's own momentum for the same reason. So a form has to be put in by hand:

$$P_{\text{mech}}^\mu = m_0 u^\mu + B^\mu, \quad (28)$$

with B^μ undetermined.

One condition is available. Since $u \cdot u = -1$ throughout, $u \cdot a = 0$, and contracting Eq. (27) with u_μ must therefore give an identity. The Lorentz force is orthogonal to u , and $u \cdot dP/d\tau = 0$ was checked in Sec. 2.1.5, and $m_0 u \cdot a = 0$. What remains is

$$u_\mu \frac{dB^\mu}{d\tau} = 0. \quad (29)$$

This says no more than that whatever force B^μ contributes must be orthogonal to the four-velocity, which is a requirement on any force term in a relativistic equation of motion. It is the only condition the tube construction imposes, though not the only one that has been imposed on B^μ in the literature: C. J. Eliezer (1947) requires in addition that $v_\lambda P_{\text{mech}}^\mu - v^\mu P_{\text{mech} \lambda}$ be a perfect differential, which is what forces the series of Sec. 5.5 to contain only even derivatives.

It is a very weak condition. One scalar equation restricts four functions, so three functions of τ remain free at every instant. Dirac's choice is the simplest solution,

$$B^\mu = k u^\mu, \quad k = \text{const}, \quad (30)$$

for which $dB^\mu/d\tau = k a^\mu$ and Eq. (29) holds automatically. Substituting Eqs. (28) and (30) into Eq. (27) with Eq. (26),

$$\left(m_0 + k + \frac{e^2}{2\epsilon c^2} \right) a^\mu = F_{\text{ext}}^\mu + \frac{2e^2}{3c^3} (\dot{a}^\mu - u^\mu a^\beta a_\beta), \quad (31)$$

and the entire content of Dirac's choice is visible in the bracket: k appears only added to m_0 and to the divergent $e^2/2\epsilon c^2$. Choosing $B^\mu = k u^\mu$ is choosing to add nothing to the dynamics. It shifts a mass, and the mass is not observable separately in any case, since only the sum

$$m = m_0 + k + \frac{e^2}{2\epsilon c^2} \quad (32)$$

appears. Holding m finite as $\epsilon \rightarrow 0$ requires $m_0 + k \rightarrow -\infty$, which should be recorded plainly. The bare mass is negatively infinite, and the observed mass is the residue of a cancellation between two quantities. Neither quantity is separately meaningful.

Two things follow, and they point in opposite directions.

The first is that the derivation has not determined the equation of motion. It has determined it *given* Eq. (30), and Eq. (30) was chosen for simplicity, not derived. Any

$$B^\mu = P_0 u^\mu + P_1 a^\mu + P_2 \dot{a}^\mu + \dots \quad (33)$$

whose derivative is orthogonal to u is equally admissible, and each choice yields a different equation of motion. It is generally of higher order, since each extra term carries another derivative. Dirac himself put it as a choice of the simplest form rather than as a result. Section 5 takes that door seriously and walks through it.

The second is that the freedom is less mysterious than the notation makes it look, because a particular choice of B^μ is nothing but a choice of what to call the particle's momentum. Suppose one posits, instead of Eq. (30),

$$P_{\text{mech}}^\mu = m_0 u^\mu + C \frac{e^2}{c^3} a^\mu \quad (34)$$

and asks what C must be if the field side is to contain only the irreversible radiated momentum. In that bookkeeping the Schott piece has been moved from the field momentum into P_{mech}^μ . Contracting the source-free balance with u_μ , the mechanical side gives $u \cdot (m_0 a + Ce^2 \dot{a}/c^3) = -Ce^2 a \cdot a/c^3$. The remaining radiative field piece contributes $-\frac{2}{3}e^2 a \cdot a/c^3$, so the two cancel only if

$$C = -\frac{2}{3}. \quad (35)$$

So orthogonality does fix the coefficient, once the *form* has been assumed. This is the same situation as before, wearing different clothes. What is worth noticing is the object it fixes. The extra piece of the momentum is $-\frac{2}{3}e^2 a^\mu/c^3$, which is the Schott momentum. It is the same quantity that appeared in Sec. 1.3.2 as the boundary term $(2e^2/3c^3) \dot{\mathbf{v}} \cdot \mathbf{v}$ that the energy-balance derivation was permitted to discard. It was discarded there by assuming periodic motion. Here it cannot be discarded, and it has to be carried as part of what the particle owns.

2.3. The equation

Absorbing the mass as in Eq. (32), Eq. (31) is the Lorentz–Dirac equation,

$$\boxed{ma^\alpha = F_{\text{ext}}^\alpha + \frac{2e^2}{3c^3} (\delta^\alpha_\beta + u^\alpha u_\beta) \dot{a}^\beta} \quad (\text{LD})$$

the projector $\delta^\alpha_\beta + u^\alpha u_\beta$ being the compact way of writing $\dot{a}^\alpha - u^\alpha a \cdot a$, and enforcing $u \cdot a = 0$ term by term.

The reaction force has two pieces and they are not the same kind of thing. The piece $\frac{2e^2}{3c^3} \dot{a}^\alpha$ is the Schott term. It is a total derivative, it changes sign with $\dot{a}$, and it is not radiation. It is momentum lent to the near field and returned. The piece $-\frac{2e^2}{3c^3} u^\alpha a \cdot a$ points along the four-velocity, never changes sign, and is exactly the Larmor power carried off, as Sec. 2.1.5 verified. Only the second is irreversible. Almost every puzzle in Sec. 3 is a puzzle about the first. The radiated power and the work done by the reaction force fail to match instant by instant. Uniform acceleration radiates while feeling no reaction at all.

It is worth stating what this derivation did and did not buy, since the same equation was very nearly written down in Sec. 1.3 by Landau and Lifshitz's two-line covariantization, Eq. (8). What is new is not the form. Covariance and $u \cdot a = 0$ fix the form almost by themselves, and the coefficient $2e^2/3c^3$ was already fixed by the nonrelativistic limit. What is new is, first, that no extended body was needed and therefore no binding stress was omitted, which is why the electromagnetic mass came out as U_0/c^2 rather than $4U_0/3c^2$. Second, the freedom left over has been made explicit rather than

hidden, in the shape of Eq. (33). A derivation that tells you exactly which of its conclusions were choices is worth more than one that reaches the same equation silently.

What the derivation did not buy is any improvement in the behaviour of the solutions. Equation (LD) still contains $\dot{a}$, so it is still of third order in the position, and Sec. 1.4 already showed what that implies. Section 3 works out the consequences.

2.4. The other route: separating the singular part



Figure 4. What the retarded and advanced potentials are, geometrically. The light cone through a field point x meets the world line twice: once in the past, at $z(u)$, and once in the future, at $z(v)$. The first gives A_{ret} , the second A_{adv} . (a) For x well off the world line the two events are far apart, and only the retarded one is causally admissible. (b) As x is brought towards the world line the two roots approach each other and both approach the same event. Both potentials diverge there, at the same rate and with the same coefficient, because both are being sourced by the same piece of the same world line; that is why the singular parts cancel in the difference $\frac{1}{2}(A_{\text{ret}} - A_{\text{adv}})$ and survive undiminished in the sum $\frac{1}{2}(A_{\text{ret}} + A_{\text{adv}})$. Equation (36) splits them where it does for this reason and no other.

There is a second way through, and it has the opposite character. Dirac’s route declines to ask what force the charge exerts on itself and audits the momentum instead. This one asks about the force after all, but subtracts, before asking, the part of the self-field that makes the question meaningless.

The difficulty being addressed is the one stated in Sec. 1: at the electron both the potential and the field tensor are infinite, so the Lorentz force expression cannot be applied to the self-field. The proposal is to split the infinite self-field into an “irrelevant” singular part, which is discarded, and a finite “effective” part, and then to apply the Lorentz force law to the effective part alone, in the ordinary way. The discarded piece is called the singular term, and the procedure is the method of separating the singular part.

The split that works is between the advanced and retarded potentials. The advanced potential A_{adv} violates causality and is not a candidate for the physical field, but it is a perfectly good solution of $\square A^\alpha = -4\pi j^\alpha$, and near the world line its singular behaviour is identical to that of A_{ret} . In the difference the singular behaviour cancels. What is kept and what is thrown away are therefore

$$A_{(R)}^\alpha = \frac{1}{2}(A_{\text{ret}}^\alpha - A_{\text{adv}}^\alpha), \quad A_{(S)}^\alpha = \frac{1}{2}(A_{\text{ret}}^\alpha + A_{\text{adv}}^\alpha), \quad (36)$$

the first being a difference of two solutions of the inhomogeneous equation and hence a solution of the homogeneous Maxwell equations, regular on the world line, with a finite Lorentz force $f^\alpha = e F^{(R)\alpha}{}_\beta u^\beta$.

That this reproduces Dirac's answer is the content of the equivalence theorem:

The motion obtained from momentum–energy conservation is the same as the motion produced by the Lorentz force of the potential $\frac{1}{2}(A_{\text{ret}} - A_{\text{adv}})$.

Part of the check is already in hand. The finite term computed in Sec. 2.1.5 is precisely e times $\frac{1}{2}(F_{\text{ret}} - F_{\text{adv}})$ evaluated on the world line, which is why the same $2e^2/3c^3$ appeared there. The agreement of the two routes is not an accident, but neither is it obvious: one of them never evaluates a field at the particle and the other does nothing else.

Proofs are given by Z. Zhang (1979) and, in retarded coordinates, by E. Poisson (2004). The version reproduced here is deferred to Sec. 4.1, and it is worth saying why it gets a section of its own rather than a footnote. Its value is not that it confirms Eq. (LD) a third time. It is that carrying the proof out makes visible something the statement conceals: that Eq. (36) is a *choice* of which combination to discard, and that the choice was made on grounds of convenience. Section 4 relaxes it, replacing Eq. (36) by a two-parameter family, and everything in that section descends from the observation that the split is not forced.

2.5. The end caps, and where the freedom in B^μ comes from

One reservation about the derivation remains to be entered, and it turns out not to be independent of the freedom found in Sec. 2.2.

The region V occupied by the tube between τ_1 and τ_2 is bounded by the wall Σ_{tube} and by two spacelike end caps $C(\tau_1)$ and $C(\tau_2)$, the discs closing the tube at each end. Conservation applies to the whole boundary, not to part of it:

$$\int_{C(\tau_2)} T^{\mu\nu} d\Sigma_\nu - \int_{C(\tau_1)} T^{\mu\nu} d\Sigma_\nu + \int_{\Sigma_{\text{tube}}} T^{\mu\nu} d\Sigma_\nu = \int_V T^{\mu\nu}{}_{,\nu} dV. \quad (37)$$

Both the calculation of Sec. 2.1 and Poisson's shorter version (E. Poisson 2004) evaluate the third term only. The 2006 text records this and does not pursue it.

The two cap integrals are not incidental. Each is the four-momentum of the field contained inside the tube at one end,

$$P_{\text{field}}^\mu(\tau) = \int_{C(\tau)} T^{\mu\nu} d\Sigma_\nu, \quad (38)$$

so what Eq. (37) says is that the wall integral measures the change in the field momentum *inside* the tube, and not the change in the particle's momentum. Those are different quantities, and separating them requires knowing how the contents of the tube divide between the particle and its field.

Which is exactly what Sec. 2.2 had to put in by hand. If Eq. (38) could be evaluated, P_{mech}^μ would follow from it rather than being posited, Eq. (28) would not be needed, and the four-vector B^μ would have no freedom left in it. The arbitrariness in B^μ is not a second defect standing alongside the missing caps. It is the missing caps, seen from the side of the equation of motion.

The caps are omitted because they diverge. On a cap the Coulomb energy density goes as e^2/r^4 against a volume element $r^2 dr$, and the momentum density, carrying one factor of $\mathbf{B} \sim 1/r$, as e^2/r^3 ; the two radial integrals

$$\int_0^\infty \frac{dr}{r^2}, \quad \int_0^\infty \frac{dr}{r} \quad (39)$$

diverge at the lower limit, the second only logarithmically but no less fatally. The field momentum inside the tube is infinite, which is why it cannot be apportioned, which is why a form for P_{mech}^μ has to be assumed.

This closes the account of Sec. 2.1 in a way worth stating plainly. The virtue of the tube construction and its cost are the same property. It works because it is insensitive to what lies inside it, and it is for that reason unable to say anything about what lies inside it. The divergence was not removed. It was moved to a place where it could be ignored, and the price of ignoring it is one undetermined four-vector.

A further consequence of the same divergence is left aside here. The stress–energy tensor of the electron’s own field curves spacetime, and a quantity that diverges on the world line curves it without bound, so the particle is not in Minkowski space and general relativity has nothing to say about the region where the problem lives. The 2006 text raises this in its Sec. 4 and builds an “effective metric” proposal on it; that material is not covered in these notes.

3. THE LORENTZ–DIRAC EQUATION: DISCUSSION AND SOLUTIONS

Equation (LD) has been derived twice and its freedoms accounted for. This section asks what it says, and the answer is that it says very nearly everything the Abraham–Lorentz equation said in Sec. 1, in covariant dress. That is not a small disappointment. Dirac’s construction repaired the derivation — no extended body, no omitted binding stress, no $4/3$ — and left the disease untouched.

3.1. One-dimensional reduction

Restrict the motion to one spatial dimension. The normalization $u^\alpha u_\alpha = -1$ reads $\dot{t}^2 - \dot{x}^2 = 1$, which is solved identically by the rapidity $q(\tau)$,

$$\dot{t} = \cosh q, \quad \dot{x} = \sinh q, \quad (40)$$

so that q carries the whole of the kinematics and the constraint is disposed of once and for all. Differentiating,

$$a^\alpha = \dot{q}(\sinh q, \cosh q), \quad \dot{a}^\alpha = \ddot{q}(\sinh q, \cosh q) + \dot{q}^2(\cosh q, \sinh q). \quad (41)$$

Two things follow at once. First $a \cdot a = \dot{q}^2$, so the invariant acceleration is just $|\dot{q}|$. Second, and more usefully, the awkward term in Eq. (LD) collapses:

$$\dot{a}^\alpha - u^\alpha a \cdot a = \ddot{q}(\sinh q, \cosh q), \quad (42)$$

because the $\dot{q}^2$ pieces of $\dot{a}^\alpha$ and of $u^\alpha a \cdot a$ are identical and cancel. Both are proportional to the same vector $(\sinh q, \cosh q)$ that carries a^α , so the two components of Eq. (LD) are not independent: they reduce to a single scalar equation,

$$\dot{q} - \tau_0 \ddot{q} = \frac{1}{m} f(\tau), \quad (43)$$

where $f = eE$ is the scalar force. In the normalization of the 2006 text, $m = e = c = 1$, one has $\tau_0 = 2/3$ and Eq. (43) is written $\dot{q} - \frac{2}{3}\ddot{q} = F^{01}$.

The reduction is worth pausing on, because it makes the structural point of Sec. 1.4 unmissable. Newton’s equation in this variable would be $m\dot{q} = f$: rapidity rate proportional to force, no more. What Eq. (43) adds is $\ddot{q}$, and it adds it with a coefficient τ_0 that is fixed, not adjustable. The equation is first order in the acceleration $\dot{q}$, which is to say third order in the position.

3.2. The free solution and the runaway

Set $f = 0$. Equation (43) becomes $\ddot{q} = \dot{q}/\tau_0$, whence

$$q = C_1 e^{\tau/\tau_0} + C_2 \quad \text{and} \quad \dot{t} = \cosh(C_1 e^{\tau/\tau_0} + C_2), \quad \dot{x} = \sinh(C_1 e^{\tau/\tau_0} + C_2). \quad (44)$$

In the 2006 normalization this is $q = C_1 e^{3\tau/2} + C_2$.

If $C_1 \neq 0$ then $q \rightarrow \infty$, and since $dx/dt = \tanh q$, the velocity approaches c . A free electron, in an inertial frame, with nothing whatever acting on it, accelerates itself to the speed of light. The covariant equation has improved on Eq. (AL) only in the sense that the velocity now saturates at c rather than growing without bound; the world line is no less absurd.

If $C_1 = 0$ then q is constant, the electron moves uniformly, and Newton's first law is recovered. This is the physical solution, and the difficulty is the one already met in Sec. 1.4: nothing selects it. The three constants of a third-order equation are fixed by three pieces of data, and position and velocity supply only two. The third is $\dot{q}(0)$, the initial acceleration, which the physical situation does not determine and which the theory offers no way of measuring; and the only value of it that does not lead to disaster must be imposed from outside.

The impulse problem sharpens this exactly as it did before. With $f/m = \kappa \delta(\tau)$ and the electron moving uniformly beforehand, integrating Eq. (43) across the pulse and solving on either side gives, for $\tau > 0$,

$$\dot{q} = -\frac{\kappa}{\tau_0} e^{\tau/\tau_0}, \quad (45)$$

so the acceleration grows without bound after the blow instead of returning to zero. The problem Sec. 1.4 found in Lorentz's equation is untouched by Dirac's. The 2006 text notes that Zhang's treatment of the single and double pulse (Z. Zhang 1979) produces solutions of the same form.

3.3. Dirac's asymptotic condition and the price paid

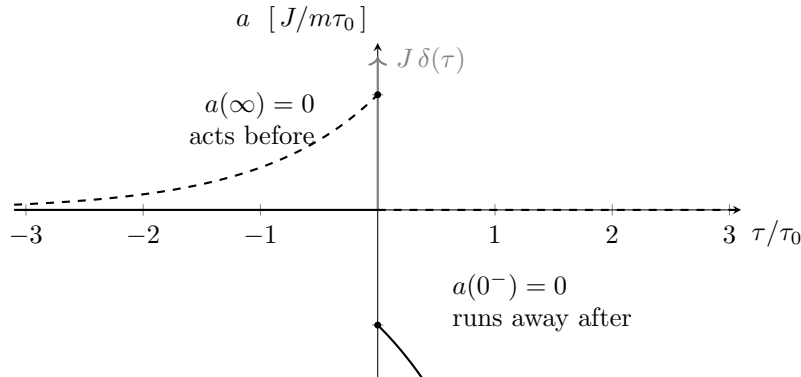


Figure 5. The choice Dirac's asymptotic condition makes. Both curves solve the same equation with the same impulse, and differ only in which end condition is imposed. Fixing the acceleration to vanish before the blow, as an initial-value problem would, gives the solid curve: nothing happens until $\tau = 0$ and the particle then runs away, as in Fig. 1(b). Fixing it to vanish at $\tau \rightarrow +\infty$ instead gives the dashed curve: the runaway is gone, the particle ends in uniform motion with exactly the velocity change J/m that the impulse should produce, and it acquires that velocity *before* the impulse arrives, over a time of order τ_0 . The two branches are reflections of each other through the origin. There is no third branch, and no choice that avoids both.

Dirac's response was not to modify the equation but to change where the data are supplied. Instead of prescribing the state at some initial instant, impose

$$\dot{q}(\tau) \longrightarrow 0 \quad \text{as} \quad \tau \rightarrow \pm\infty, \quad (46)$$

that is, require the acceleration to die at both ends of the world line. For the impulse problem this fixes the free constant uniquely, and the solution turns out to be

$$\dot{q} = \frac{\kappa}{\tau_0} e^{\tau/\tau_0} \quad (\tau < 0), \quad \dot{q} = 0 \quad (\tau > 0). \quad (47)$$

This works, and it works well. The runaway is gone. The velocity change delivered is $\int_{-\infty}^0 \dot{q} d\tau = \kappa$, exactly what an impulse of that strength should produce, so the final state is the correct one. Compared with Eq. (45) the whole pathology has been reflected through $\tau = 0$, as Fig. 5 shows.

And reflecting it through $\tau = 0$ is precisely the trouble. The electron is accelerating at $\tau < 0$. It begins to move before the force reaches it, on a timescale τ_0 , and it begins to move in the direction the force will push it, by the amount the force will deliver. It anticipates. Two objections follow, and the 2006 text states both.

The first is procedural. The condition $\dot{q}(0)$ is no longer required — it has been replaced by conditions at $\tau = \pm\infty$. But the motions one actually wants to describe occupy finite stretches of time, and it is a strange theory in which settling what an electron does over a microsecond requires knowing what it will be doing infinitely far in the future. The initial-value problem has not been solved; it has been exchanged for a boundary-value problem whose boundaries are not accessible.

The second is physical. Pre-acceleration is a violation of causality of a particularly naked kind: the effect is not merely correlated with the cause, it precedes it, and it precedes it by an amount that carries information about the cause. In defence one may say that $\tau_0 \sim 10^{-24}$ s is far below any timescale on which classical electrodynamics could be trusted, and that is true; but it is a statement about where the theory should be abandoned, not a resolution of the problem inside it.

What Eq. (46) buys, then, is criterion 2 of Sec. 1.5, and what it spends is criterion 3. That trade is the one fixed feature of this subject, and it will be worth remembering when Secs. 4 and 5 try to arrange a better bargain.

3.4. Energy balance and the Schott term

The reaction force in Eq. (LD) has two pieces, and Sec. 2.3 distinguished them by their character. Their effect on the energy budget is what makes the distinction matter. Care is needed here, because a four-force is always orthogonal to u^α . Contracting it with u_α therefore gives zero identically. It is not a power.

To speak about energy, choose an inertial observer with unit time direction n^α . The energy associated with a four-vector P^α is $E = -P \cdot n$. Write the reaction four-force as $R^\alpha = m\tau_0(\dot{a}^\alpha - u^\alpha a \cdot a)$. Its first piece is a total proper-time derivative,

$$m\tau_0 \dot{a}^\alpha = \frac{d}{d\tau} (m\tau_0 a^\alpha), \quad (48)$$

and the quantity being differentiated is the Schott momentum already met in Sec. 2.2. Its second piece, $-m\tau_0 u^\alpha a \cdot a$, is the radiated four-momentum, $-P_{\text{rad}} u^\alpha / c^2$ with $P_{\text{rad}} = (2e^2/3c^3) a \cdot a$ the Larmor

power in the instantaneous rest frame. The energy balance over an interval is therefore

$$\underbrace{\int_{\tau_1}^{\tau_2} -R \cdot n \, d\tau}_{\text{energy supplied by the reaction}} = \underbrace{\left[-m\tau_0 a \cdot n \right]_{\tau_1}^{\tau_2}}_{\text{change in Schott energy}} - \underbrace{\int_{\tau_1}^{\tau_2} P_{\text{rad}}(-u \cdot n) \, d\tau}_{\text{radiated energy seen by } n}, \quad (49)$$

which is Eq. (6) again, now written with an observer specified and with the term that Sec. 1.3.2 was allowed to discard carried explicitly.

The radiated power and the work done by the reaction force therefore do not match instant by instant. They match only when the Schott term contributes nothing over the interval, which happens for periodic motion, or when the motion begins and ends with the same value of $a \cdot n$. This is the same condition, and the same evasion, as in Sec. 1.3.2. The difference is that there it was a licence to simplify a derivation, and here it is a statement about what the equation actually asserts. Between one instant and the next, the energy the particle loses and the energy the field carries away are not the same quantity. The discrepancy is stored in, and returned by, the near field.

3.5. Hyperbolic motion

The sharpest form of the difficulty is uniformly accelerated motion. Take

$$z^\alpha(\tau) = \frac{1}{\alpha} (\sinh \alpha\tau, \cosh \alpha\tau), \quad (50)$$

a hyperbola in spacetime, with constant proper acceleration α . In the rapidity variable this is simply $q = \alpha\tau$, so $\dot{q} = \alpha$ and $\ddot{q} = 0$. By Eq. (42) the reaction force vanishes identically,

$$\dot{a}^\alpha - u^\alpha a \cdot a = 0, \quad (51)$$

which is easily checked directly, since $\dot{a}^\alpha = \alpha^2 u^\alpha$ and $a \cdot a = \alpha^2$.

So Eq. (LD) says that a uniformly accelerated charge feels no radiation reaction at all, and follows exactly the world line an uncharged particle of the same mass would follow under the same applied force. Yet $a \cdot a = \alpha^2 \neq 0$, so the Larmor formula says it radiates, steadily and forever, at the rate $(2e^2/3c^3)\alpha^2$. Something is paying for that radiation and it is not the applied force, which is doing exactly the work required to produce the motion and no more.

Equation (49) says where the accounting goes. In the inertial frame with $n^\alpha = (1, 0)$, the Schott energy is $E_{\text{Schott}} = -m\tau_0 a \cdot n = m\tau_0 \alpha \sinh \alpha\tau$. Its change over a finite interval is exactly $\int P_{\text{rad}}(-u \cdot n) d\tau$. The reaction force vanishes, so the left side of Eq. (49) is zero, and the radiated energy is paid for by the change in the Schott energy of the near field.

This is a consistent account, but it is not a comfortable one. The result depends on a near-field energy that grows without bound for eternal hyperbolic motion, and it ties the energy emitted in a finite interval to the history of a solution that has existed forever. The point to carry forward is not that hyperbolic motion is a contradiction. It has been discussed for a century and there are consistent ways of reading it. The point is that the trouble sits, once again, in the Schott term. Every awkward feature of Eq. (LD) traces back to the same $\dot{a}^\alpha$: the third order and the runaway, the pre-acceleration that removing the runaway costs, the failure of the energy budget to close instant by instant, and now a radiating particle with nothing pushing back on it.

3.6. Scorecard, and the fork in the road

Against the three criteria of Sec. 1.5:

1. **Solvable as an ordinary initial-value problem?** No. The equation is third order, so it needs three pieces of data, and the physical situation supplies two. The third is the initial acceleration, and every value of it but one leads to a runaway.
2. **Stable under a small disturbance?** Not as an initial-value problem: Eq. (45) runs away. Yes, if Eq. (46) is imposed instead, but then the problem is no longer an ordinary initial-value problem.
3. **Conservation and causality?** Causality fails as soon as Eq. (46) is imposed, by Eq. (47). Energy is conserved, but only in the integrated sense of Eq. (49), and Sec. 3.5 shows how uncomfortable that can get.

The pattern is worth stating in one line, because it is what the rest of these notes is a response to. As an initial-value problem, LD keeps the equation local and causal but is unstable. With Dirac's asymptotic condition it becomes stable, but only by becoming a boundary-value problem with pre-acceleration. No choice of boundary condition satisfies all three criteria, because the difficulty is not in the boundary conditions. It is in the $\dot{a}^\alpha$.

Two ways of attacking that term present themselves, and they are the subjects of the two sections that follow.

- *Sideways.* Keep the structure of the reaction term and open up its coefficient. Section 4 does this, arriving at it from the observation of Sec. 2.4 that the retarded–advanced split of Eq. (36) was a choice.
- *Upward.* Keep the coefficient and raise the order of the equation by enlarging B^μ beyond Dirac's simplest choice, along the lines left open by Eq. (33). Section 5 does this.

This contrast is the spine of the notes. One of the two routes will turn out to survive its own motivation and the other will not, but that is not visible yet, and both are pursued on their own terms.

4. SPLITTING OFF THE SINGULAR PART: THE λ – κ FAMILY

This is the sideways branch of the fork of Sec. 3.6: keep the order of the equation and open up the coefficient of the reaction term. The door was left open in Sec. 2.4. There the self-field was split into a singular part to be discarded and an effective part to be kept,

$$A_{(R)}^\alpha = \frac{1}{2} (A_{\text{ret}}^\alpha - A_{\text{adv}}^\alpha), \quad A_{(S)}^\alpha = \frac{1}{2} (A_{\text{ret}}^\alpha + A_{\text{adv}}^\alpha),$$

the decomposition being Dirac's (P. A. M. Dirac 1938), and the modern form of the same idea being the singular–regular split of S. Detweiler & B. F. Whiting (2003). It was said in Sec. 2.4 that this split is a choice made on grounds of convenience rather than a result. This section makes good on that, by first proving that the choice reproduces Dirac's equation (Sec. 4.1), then asking whether the discarded part is really inert (Sec. 4.2), and then relaxing the split into a two-parameter family and following where the parameters lead.

Of the two branches of the fork this is the one to watch. Whatever becomes of the argument that motivates it — and Sec. 4.2 will have to be read with some care — the equation it produces is Eq. (LD) with the coefficient of the reaction term replaced by a free parameter, and that object has outlived everything else in D. Zhang (2006).

4.1. The equivalence theorem

The motion obtained from momentum–energy conservation is the same as the motion produced by the Lorentz force of the potential $\frac{1}{2}(A_{\text{ret}} - A_{\text{adv}})$.

The theorem is not new. Proofs are given by Z. Zhang (1979) and, in retarded coordinates, by E. Poisson (2004), and neither is short. The one reproduced below is the elementary argument of D. Zhang (2006), given there precisely because the standard routes are laborious. It is worth setting out not only because it is quicker but because of what becomes visible while it is being carried out.

Start from the retarded potential of the charge (see, e.g., J. D. Jackson 1999, ch. 14),

$$A_{\text{ret}}^{\alpha}(t, \mathbf{x}) = \int \frac{j^{\alpha}(t - |\mathbf{x} - \mathbf{x}'|, \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x', \quad (52)$$

and expand the source about the present time rather than the retarded one. Writing $R = |\mathbf{x} - \mathbf{x}'|$ and using $j^{\alpha}(t - R) = \sum_l \frac{(-R)^l}{l!} \partial_t^l j^{\alpha}(t)$,

$$A_{\text{ret}}^{\alpha} = \int \sum_{l=0}^{\infty} \frac{(-1)^l}{l!} R^{l-1} \partial_t^l j^{\alpha}(t, \mathbf{x}') d^3x'. \quad (53)$$

The advanced potential is the same integral with $t + R$ in place of $t - R$, so its expansion is identical except that the factor $(-1)^l$ is absent,

$$A_{\text{adv}}^{\alpha} = \int \sum_{l=0}^{\infty} \frac{1}{l!} R^{l-1} \partial_t^l j^{\alpha}(t, \mathbf{x}') d^3x'. \quad (54)$$

Everything now follows from the parity of l . In the half-difference the coefficient of the l th term is $\frac{1}{2}[(-1)^l - 1]$, which is zero for every even l and -1 for every odd one: *the even terms cancel and the odd terms survive*. Inserting the point-charge source, $j^0 = e \delta(\mathbf{x} - \mathbf{z}(t))$ and $\mathbf{j} = e \mathbf{v} \delta(\mathbf{x} - \mathbf{z}(t))$, and carrying out the $\mathbf{x}'$ integration against the delta functions,

$$\mathbf{A}_{rr} = - \sum_{l \text{ odd}} \frac{1}{l!} \frac{\partial^l}{\partial t^l} [\mathbf{v}(t) |\mathbf{x} - \mathbf{z}(t)|^{l-1}], \quad \Phi_{rr} = - \sum_{l \text{ odd}} \frac{1}{l!} \frac{\partial^l}{\partial t^l} |\mathbf{x} - \mathbf{z}(t)|^{l-1}, \quad (55)$$

writing A_{rr} for $\frac{1}{2}(A_{\text{ret}} - A_{\text{adv}})$.

The regularity of A_{rr} can now be read off, and it is not luck. (The argument in the next two sentences is not in D. Zhang 2006, which establishes finiteness instead by noting that a difference of two solutions of the inhomogeneous equation solves the homogeneous one. The conclusion is the same; the route below makes it visible one step earlier, and generalizes.) The only term of Eqs. (53) and (54) carrying a negative power of R is $l = 0$, which is the Coulomb term; and $l = 0$ is even, so it is exactly the term the half-difference removes. Every surviving term has $l \geq 1$, hence a non-negative

power R^{l-1} , and nothing in Eq. (55) blows up as $\mathbf{x} \rightarrow \mathbf{z}(t)$. This is the algebraic form of what Fig. 4 showed geometrically: the two potentials are sourced by the same piece of the same world line in the same way and differ only in the sign of the retardation, so their singular parts are identical and cancel in the difference.

The rest is computation. With A_{rr} regular one may take $\mathbf{E} = -\nabla\Phi_{rr} - \partial_t\mathbf{A}_{rr}$ and $\mathbf{B} = \nabla \times \mathbf{A}_{rr}$, evaluate them on the world line, and form $\mathbf{F} = e(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. Carrying out the differentiations — the $l = 1$ and $l = 3$ terms of Eq. (55) are the ones that contribute — gives the three-dimensional form of the Lorentz–Dirac equation, which is the content of Appendices A.1 and A.2 of D. Zhang (2006). The theorem is proved, and the equation of Sec. 2.3 — obtained there from momentum–energy conservation with no field ever evaluated at the particle — has been recovered from a Lorentz force with a particular potential.

What the proof exposes is the thing to carry forward, and this too goes beyond D. Zhang (2006), which arrives at the same condition from the field-theoretic side rather than from the expansion. The cancellation used no property of $\frac{1}{2}(A_{\text{ret}} - A_{\text{adv}})$ beyond the relative sign of its two pieces. For a general combination

$$A^\alpha = \lambda A_{\text{ret}}^\alpha + \kappa A_{\text{adv}}^\alpha, \quad (56)$$

the $l = 0$ term comes with the coefficient $\lambda + \kappa$, so

$$A^\alpha \text{ is regular on the world line} \iff \lambda + \kappa = 0. \quad (57)$$

This is the same statement as “the retained part satisfies the homogeneous Maxwell equations”, since $\lambda A_{\text{ret}}^\alpha + \kappa A_{\text{adv}}^\alpha$ has source $(\lambda + \kappa)j^\alpha$; the parity argument simply makes it visible one step earlier. And it says that $\frac{1}{2}(A_{\text{ret}} - A_{\text{adv}})$ is not singled out. It is one conveniently normalized point on a line of equally regular choices, and Sec. 4.4 is what happens if one moves along that line.

4.2. Does the singular part act?

The split of Eq. (36) is legitimate only if the part thrown away does nothing. That is a substantive claim and D. Zhang (2006) examines it rather than assuming it. The examination is the load-bearing step of this section, and it is also the step most in need of checking, so both halves of that are set out here.

Begin with the difficulty that makes the claim non-trivial. Computing the field of $A_{(S)}$ in retarded coordinates and contracting it with the four-velocity gives a four-force whose leading behaviour is

$$f^\alpha = e F^{(S)\alpha}{}_\beta u^\beta = e^2 \left[\frac{k^\alpha - u^\alpha}{r^2} + O\left(\frac{1}{r}\right) \right], \quad (58)$$

and k^α does not approach u^α as $r \rightarrow 0$: the angle between them is fixed along each generator. So Eq. (58) is an infinite force, going as $1/r^2$, with nothing to balance it. This is the difficulty of Sec. 1 returning in its original form. To make the point picture work one has to delete an unbalanced infinity, and deleting it is not something the theory licenses.

What licenses it, in the modern treatment, is a separate principle: the singular field exerts no force on the particle, all of the self-force coming from the regular field. This is the Detweiler–Whiting principle (S. Detweiler & B. F. Whiting 2003), and D. Zhang (2006) invokes it by name, from Poisson’s review (E. Poisson 2004), while describing it frankly as a practical rule standing in for a principle that is missing.

The question is then a sharp one. *Is the Detweiler–Whiting singular field the same object as $\frac{1}{2}(A_{\text{ret}} + A_{\text{adv}})$?* If it is, the split of Eq. (36) is forced and there is nothing to relax. The 2006 text answers no, and the answer is a computation. Building the stress–energy tensor of the symmetric combination,

$$T_{(S)}^{\alpha\beta} = \frac{1}{4\pi} \left[F^{(S)\alpha}{}_{\delta} F^{(S)\delta\beta} - \frac{1}{4} \eta^{\alpha\beta} F_{\mu\rho}^{(S)} F^{\mu\rho}_{(S)} \right], \quad (59)$$

and integrating its flux in the manner of Sec. 2.1 gives, for a unit charge, Eq. (1.2.17) of D. Zhang (2006):

$$\frac{dP^\alpha}{d\tau} = \frac{a^\alpha}{2r} + \frac{2}{3} \dot{a}^\alpha. \quad (60)$$

The first term is the familiar divergent piece proportional to the acceleration, and is absorbed into the mass as in Sec. 2.2. The second is finite and does not vanish. Read as the effect of $A_{(S)}$ on the particle, it says that the symmetric combination is *not* inert, hence is not the Detweiler–Whiting singular field, hence that Eq. (36) is one choice among others. That is the licence Sec. 4.3 uses.

Now the other half. Equation (60) should not be relied on without an independent recomputation, for two reasons.

The first is that it stands against the literature. In flat spacetime the Detweiler–Whiting singular field *is* the half-sum, and the theorem (S. Detweiler & B. F. Whiting 2003) is precisely that it exerts no force. Both statements cannot be true. The 2006 text is aware of the principle and is proposing that the identification fails; the modern position is that it does not.

The second is that the quantity computed may not be the quantity wanted. The force on the particle follows from the balance of the *total* field, and

$$T[F_{\text{ret}}] = T[F_{(S)}] + T[F_{(R)}] + 2T[F_{(S)}, F_{(R)}], \quad (61)$$

so the flux of the singular field’s own stress tensor omits the cross terms. “The momentum carried by the singular field” and “the force the singular field exerts” are different quantities, and Eq. (60) computes the first. Whether the $\frac{2}{3}\dot{a}^\alpha$ survives once Eq. (61) is treated in full, or whether it is an artefact of the algebra in Appendix A.3 of D. Zhang (2006), is not settled here.

What can be said without settling it is stated in Sec. 4.5, and it is worth flagging now so that the reader is not left waiting: the family constructed in the next two subsections does not depend on Eq. (60) for its usefulness, only for its original motivation.

4.3. The two-parameter family

Relax the form of the discarded part, but keep the requirement that what remains be built from the same two potentials. Writing the retained field as in Eq. (56),

$$A^\alpha = \lambda A_{\text{ret}}^\alpha + \kappa A_{\text{adv}}^\alpha,$$

the parity decomposition of Sec. 4.1 does the work immediately. The l th term of A_{ret} carries $(-1)^l$ and that of A_{adv} carries 1, so the combination is

$$A^\alpha = (\kappa - \lambda) \sum_{l \text{ odd}} \frac{1}{l!} \partial_t^l[\cdot] + (\lambda + \kappa) \sum_{l \text{ even}} \frac{1}{l!} \partial_t^l[\cdot], \quad (62)$$

the brackets standing for the same quantities as in Eq. (55). The odd series is the regular one; the even series is the one containing $l = 0$ and hence the whole of the singularity.

Two conditions on (λ, κ) then suggest themselves, and D. Zhang (2006) takes both.

1. **Branch (a)**, $\lambda + \kappa = 0$. The even series drops out entirely, so the retained field is regular on the world line. This is Eq. (57), and equivalently the statement that the retained field satisfies the homogeneous Maxwell equations. Section 4.4.
2. **Branch (b)**, $\lambda - \kappa = 1$. The *discarded* part is $(1 - \lambda)A_{\text{ret}}^\alpha - \kappa A_{\text{adv}}^\alpha$, and this condition makes retarded and advanced enter it with equal weight. The retained field then keeps a singular part. Section 4.5.

These are two lines in the (λ, κ) plane and they meet at one point,

$$\lambda = \frac{1}{2}, \quad \kappa = -\frac{1}{2}, \quad (63)$$

which is Dirac's choice, Eq. (36). So the standard split is the unique member of the family satisfying both conditions at once. That is worth stating, because it explains why $\frac{1}{2}(A_{\text{ret}} - A_{\text{adv}})$ looks canonical: it is distinguished, just not uniquely by either requirement on its own.

A third line through the same plane is worth recording, because it was drawn first and it arrives at the same place. C. J. Eliezer (1947) takes the field of the electron to be $F_{\text{ret}} + k(F_{\text{ret}} - F_{\text{adv}})$, that is $\lambda = 1 + k$ and $\kappa = -k$, so that $\lambda + \kappa = 1$: the retained field keeps the full source rather than none of it. The singular part is then disposed of not by cancelling it but by the stipulation of Sec. 4.2 that it exerts no force, and what is left acting is $(2k + 1)$ times the Dirac field. Eliezer's resulting equation of motion, his Eq. (61), is Eq. (LD) with its reaction term multiplied by $2k + 1$. The two routes could hardly look less alike — one kills the singular part, the other keeps it and declares it inert — and they produce the same one-parameter family of equations.

Branch (a) can be settled here, before any equation of motion is solved. Setting $\lambda = -\kappa$ in Eq. (62) leaves $\kappa - \lambda = 2\kappa$, so

$$A^\alpha|_{\lambda+\kappa=0} = -2\kappa A_{rr}^\alpha, \quad (64)$$

with A_{rr} the Dirac field of Sec. 4.1. The retained field on this branch is nothing but Dirac's, rescaled by -2κ . Field, force and reaction term are all linear in the potential, so the reaction term of Eq. (LD) is rescaled by the same factor and nothing else changes:

$$\frac{2e^2}{3} \longrightarrow -\frac{4\kappa e^2}{3}, \quad (65)$$

which is Eq. (1.5.1) of D. Zhang (2006), recovered without solving anything. Setting $\kappa = -\frac{1}{2}$ returns the coefficient $2e^2/3$ and with it Eq. (LD), as Eq. (63) requires.

The whole of branch (a) is therefore the Lorentz–Dirac equation with one number changed. Section 4.4 has only to work out what changing it does to the solutions. Branch (b) admits no such shortcut, since the retained field keeps its singular part and the rescaling argument does not apply; it has to be handled numerically, in Sec. 4.5.

4.4. Branch (a): $\lambda + \kappa = 0$

Section 4.3 already established what this branch is: the Lorentz–Dirac equation with its reaction coefficient rescaled by -2κ ,

$$ma^\alpha = F_{\text{ext}}^\alpha - \frac{4\kappa}{3}e^2 (\dot{a}^\alpha - a^\beta a_\beta u^\alpha), \quad (66)$$

with $\kappa = -\frac{1}{2}$ returning Eq. (LD). What remains is to find out what the rescaling does, and for the free particle in one dimension that can be answered in closed form.

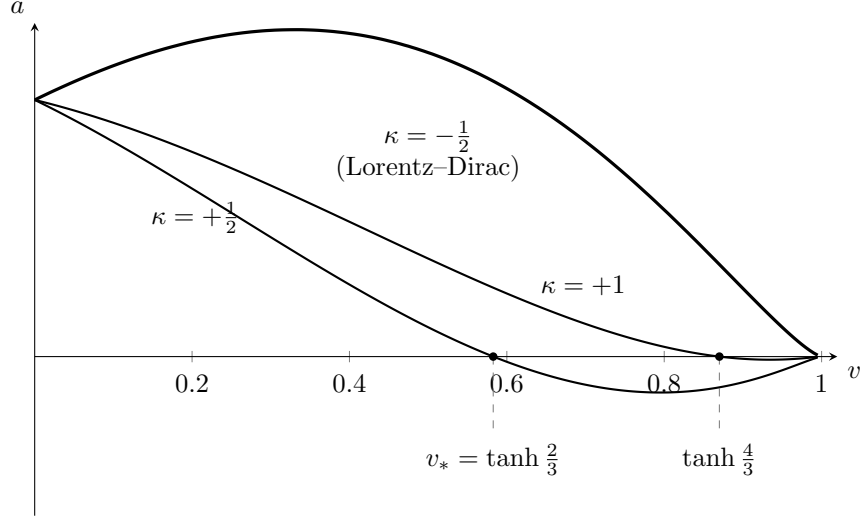


Figure 6. The phase portrait of branch (a), Eq. (68), for $C = 1$. The sign of κ decides everything. For $\kappa < 0$, which includes the Lorentz–Dirac value $\kappa = -\frac{1}{2}$, the bracket grows with v and the curve never returns to the axis: the acceleration is positive at every speed, so nothing stops the particle short of c , and this is the runaway of Sec. 3.2 seen in the (v, a) plane. For $\kappa > 0$ the bracket decreases, the curve must cross the axis, and the crossing at $v_* = \tanh(4\kappa C/3)$ is an attractor: below it the acceleration is positive and above it negative, so a particle released anywhere returns to v_* . The terminal speed is then fixed by C , which is fixed by the applied impulse — force determines velocity rather than acceleration, which is why D. Zhang (2006) calls this Aristotle motion. Only $v > 0$ is shown; the portrait is symmetric under $(v, a, C) \rightarrow (-v, -a, -C)$.

Write $a = \dot{v} = p(v)$, so that $\ddot{v} = p dp/dv$ and the equation of motion becomes first order in the (v, a) plane. With no external force Eq. (66) reduces to

$$\frac{dp}{dv} + \frac{3v}{1-v^2} p = -\frac{3}{4\kappa} \sqrt{1-v^2}, \quad (67)$$

a linear equation whose integrating factor is $(1-v^2)^{-3/2}$, and which therefore integrates at once to

$$a = \left[-\frac{3}{8\kappa} \ln \frac{1+v}{1-v} + C \right] (1-v^2)^{3/2}. \quad (68)$$

This is Eq. (1.2.26) of D. Zhang (2006). The constant C is fixed by the state the particle is put in — for a particle kicked from rest, by the size of the kick — and κ is the parameter of the family. Everything about this branch can be read off the one curve, and it is drawn in Fig. 6.

The sign of κ decides the whole question, and it decides it by elementary means. As v runs from 0 to 1 the logarithm runs from 0 to $+\infty$. So for $\kappa < 0$ the bracket *increases* without bound, the curve stays above the axis at every speed, and the acceleration is positive everywhere: nothing brings the particle to rest relative to its own motion and it proceeds to c . That is the runaway of Sec. 3.2, and since $\kappa = -\frac{1}{2}$ lies in this range it is also the statement that Eq. (LD) belongs to the bad half of the family.

For $\kappa > 0$ the bracket *decreases*, from C at $v = 0$ towards $-\infty$. It must therefore vanish somewhere, and where it vanishes the acceleration does too. Setting the bracket to zero and using $\ln \frac{1+v}{1-v} =$

$2 \operatorname{artanh} v$,

$$v_* = \tanh\left(\frac{4\kappa C}{3}\right). \quad (69)$$

Below v_* the acceleration is positive and above it negative, so v_* is not merely a zero but an attractor: a particle released at any speed is carried to it. D. Zhang (2006) identifies the crossing and reads it as the point of re-equilibration; Eq. (69) writes it out.

That is a genuine damping. A particle disturbed from uniform motion settles back to uniform motion, which is what criterion 2 of Sec. 1.5 asks for and what nothing in Secs. 3 or 5 delivers.

It also has a peculiar character, and the 2006 text's name for it is the right one. The speed the particle ends at is set by C , and C is set by the force that was applied. *The force determines the velocity, not the acceleration.* That is Aristotle's picture of motion rather than Newton's, and Eq. (69) is its quantitative form; hence *Aristotle motion*. It is worth being clear that this is not a slip. It is what a damping term of this sign does: the reaction opposes the motion at every speed until the opposition balances the applied force, exactly as a body in a viscous medium reaches a terminal velocity, and Aristotle's mistake was to suppose that the world is always in that regime.

Two things should be entered against this branch before it is left.

The first is that κ is not determined. It is a free parameter, and the theory says only that it must be positive for the behaviour above. The 2006 text is explicit that it would have to be fixed by experiment, which is what its Sec. 5 is about and is not covered in these notes.

The second is more serious. It is not raised in the 2006 text, but it was raised in 1947, in one sentence, by C. J. Eliezer (1947): in the notation of his Eq. (61) a negative $2k + 1$ turns what corresponds to self-acceleration into self-retardation, and *what corresponds to emission of radiation into absorption of radiation*. Turning κ positive reverses the sign of the reaction term in Eq. (66). The term whose sign was reversed is the one identified in Sec. 2.3 as the radiated four-momentum, $-P_{\text{rad}} u^\alpha / c^2$ with $P_{\text{rad}} \geq 0$ by the Larmor formula. Reversing it does not merely damp the motion; it turns radiative loss into radiative gain, and criterion 3 of Sec. 1.5 was written down with exactly this kind of move in mind. No energy budget for Eq. (66) appears anywhere in the 2006 text, and it should be computed before the $\kappa > 0$ branch is taken to have solved anything. The runaway is certainly gone. Where the energy comes from is an open question.

4.5. Branch (b), and what the family amounts to

Branch (b) can be dealt with quickly, because nothing later depends on it. Setting $\lambda = 1 + \kappa$ leaves the retained field with a singular part, so the rescaling argument of Eq. (64) does not apply and there is no closed-form portrait; D. Zhang (2006) reduces the equation of motion and integrates it numerically with κ as the parameter. The result is one watershed and two regimes. For $\kappa > -1$, of either sign, the phase curve is convex and the electron self-accelerates much as under Eq. (LD); larger κ merely lowers the acceleration without changing its sign. At $\kappa = -1$ the portrait changes character. For $\kappa < -1$ a critical velocity v_c appears, and it is a different object from the attractor of Sec. 4.4: a particle starting below v_c remains below it for ever, approaching it asymptotically, while one starting above it decelerates stably.

One obstruction should be recorded, since it is the reason this branch is not worth pursuing further as it stands. Taking it beyond one dimension requires $\mathbf{n} \cdot \mathbf{v}$ at the position of the charge itself, where $\mathbf{n}$ is not defined. D. Zhang (2006) suggests replacing it by its angular average $\frac{1}{3}v$ and does not carry that through. Until something is done about it, branch (b) exists only in one dimension.

κ	-1.5	-2	-3	-500	-5000
v_c	0.3473	0.2261	0.1341	6.67×10^{-3}	6.67×10^{-4}

Table 2. The critical velocity of branch (b), from [D. Zhang \(2006\)](#). The two largest values give $|\kappa| v_c = 3.34$ apiece, so v_c falls off as $1/|\kappa|$ once $|\kappa|$ is large.

What the section as a whole amounts to is branch (a), and it amounts to one sentence. Equation (66) is Eq. (LD) with the coefficient of the radiation-reaction term replaced by a free parameter. Everything else — the two potentials, the parity argument, the two-parameter family, the condition $\lambda + \kappa = 0$ — is scaffolding that produces that one modification and is then dispensable.

This is worth separating carefully from the argument that motivated it. Section 4.2 rested the case for opening the split on Eq. (60), and gave reasons to doubt it: if the singular field exerts no force, as the Detweiler–Whiting principle asserts, then there was never any licence to relax Eq. (36) and the derivation collapses. But the equation it produced does not collapse with it. Written in a form that does not mention λ or κ at all,

$$ma^\alpha = F_{\text{ext}}^\alpha + \zeta \tau_0 m (\dot{a}^\alpha - a^\beta a_\beta u^\alpha), \quad \zeta = -2\kappa, \quad (70)$$

with $\zeta = 1$ the standard value and $\zeta = 1$ alone recovering Eq. (LD). This is Eliezer’s Eq. (61) again, with $\zeta = 2k + 1$; the parametrization is older than the derivation given for it here. This is a one-parameter deformation of the Lorentz–Dirac equation, and it stands on its own whatever becomes of the reasoning that suggested it. A derivation can be wrong and still have found something.

The virtue of Eq. (70) is that ζ is a number that data can constrain, which is more than can be said for anything else in these notes. The sign question of Sec. 4.4 is now the statement that $\zeta < 0$ reverses the radiated term, and the Aristotle branch is the whole half-line $\zeta < 0$. What the existing measurements have to say about ζ , and how tightly they bound it, is a separate matter and is not taken up here.

5. THE ELIEZER-TYPE SERIES

This is the upward branch of the fork of Sec. 3.6: keep the coefficient of the reaction term and raise the order of the equation. The door was left open in Sec. 2.2, where the tube calculation was found to fix the particle’s own four-momentum only through Eq. (29), and Dirac closed it with the simplest choice.

The fork itself is not an invention of these notes, nor of [D. Zhang \(2006\)](#). It is stated at the head of the section [C. J. Eliezer \(1947\)](#) devotes to the problem: there are two ways to modify the Lorentz–Dirac equation, one being to keep the field of the electron equal to its retarded field and introduce derivatives of the velocity higher than the second, the other to abandon the purely retarded field and work with a combination of retarded and advanced. Those are Secs. 5 and 4 respectively. The 2006 text pursues both without recording that the choice between them had been posed sixty years earlier.

One warning before anything else, since two different equations carry the same name. What the literature calls the Eliezer, or Eliezer–Ford–O’Connell, equation ([C. J. Eliezer 1948](#); [G. W. Ford & R. F. O’Connell 1991](#)) is obtained by *reducing* the order of Eq. (LD), is free of runaway solutions, and is a live candidate in present-day radiation-reaction experiments. What follows is the opposite

operation, carried out on an ansatz from the earlier review (C. J. Eliezer 1947). The two sit on opposite sides of the fork just described, and reading the conclusion of this section as a verdict on the first would invert it exactly. Below, “the Eliezer equation” means the second.

5.1. The Eliezer equation

Return to the balance of Sec. 2.2. Written in Eliezer’s notation the equation of motion before any choice is made is

$$\dot{A}_\mu - \frac{2}{3}e^2 \left(\frac{d\dot{v}_\mu}{d\tau} + \dot{v}^2 v_\mu \right) = e v_\sigma F_\mu^\sigma{}_{\text{ext}}, \quad (71)$$

where A_μ is the particle’s own four-momentum, the P_{mech}^μ of Eq. (28). Taking $A_\mu = mv_\mu$ gives Eq. (LD); taking $A_\mu = 0$ describes a particle of zero rest mass. The question is what else A_μ may be.

Two conditions restrict it. The first is Eq. (29), $v \cdot \dot{A} = 0$. The second, which Sec. 2.2 did not impose, is that $v_\lambda A_\mu - v_\mu A_\lambda$ be a perfect differential. Together they are much stronger than either alone. Trying the obvious two-term form,

$$A_\mu = P v_\mu + Q \dot{v}_\mu, \quad (72)$$

C. J. Eliezer (1946) showed that no choice of P and Q satisfies both: the form is simply not available. The shortest admissible extension carries one more derivative,

$$A_\mu = P v_\mu + Q \dot{v}_\mu + R \frac{d\dot{v}_\mu}{d\tau}, \quad (73)$$

and imposing the two conditions on it fixes

$$P = \text{const} - \frac{3}{2}R \dot{v}^2, \quad Q = 0, \quad R = \text{const}. \quad (74)$$

That $Q = 0$ deserves emphasis. The coefficient of $\dot{v}_\mu$ is not small or negligible; it is forbidden. So the admissible A_μ is

$$A_\mu = m v_\mu + k \left(\frac{2}{3} \frac{d\dot{v}_\mu}{d\tau} + \dot{v}^2 v_\mu \right), \quad (75)$$

with m and k constants, and substituting into Eq. (71) gives the equation this section is about,

$$\boxed{m \dot{v}_\mu - \frac{2}{3}e^2 \left(\frac{d\dot{v}_\mu}{d\tau} + \dot{v}^2 v_\mu \right) + k \frac{d}{d\tau} \left(\frac{2}{3} \frac{d\dot{v}_\mu}{d\tau} + \dot{v}^2 v_\mu \right) = e v_\nu F_\mu^\nu} \quad (\text{E})$$

the Eliezer equation, and Eq. (1.3.4) of D. Zhang (2006).

Set beside Eq. (LD) the difference is a single term. Everything but the k term *is* the Lorentz–Dirac equation, so Eq. (E) reduces to it at $k = 0$, and the whole of the modification is the proper-time derivative of the quantity whose undifferentiated form is already there. That costs one order: Eq. (LD) is third order in the position and Eq. (E) is fourth. It is the smallest step that can be taken along this branch, and Sec. 5.2 is what it buys.

Two remarks about the constants, both from C. J. Eliezer (1947) and neither in the 2006 text.

First, the exclusion of $\dot{v}_\mu$ is the same fact that Sec. 5.5 obtains by a different route. There the chain of identities from $v^\mu v_\mu = -1$ was used to trade odd derivatives for lower ones, leaving only even terms

in the series; here the two conditions on A_μ kill the odd term directly. The two arguments meet at $Q = 0$.

Second, and more consequential, k is not free. A classical point electron has two constants, m and e , and no others, so either k describes some further property of the electron or it is expressible in terms of those two. Eliezer records that he had hoped for the first, as a way of getting Planck's constant into a classical theory, and that it did not work. Dimensionally k is a mass times a length squared, that is e^4/m , so

$$k = \sigma \frac{e^4}{m}, \quad (76)$$

with σ a pure number to be determined by experiment. The family therefore has one free parameter and not two, which will matter in Sec. 5.6.

5.2. The free electron, solved

Take Eq. (E) with no external field and in the non-relativistic limit, where it reduces to a linear equation for the three-velocity,

$$\frac{d^3 \mathbf{V}}{dt^3} - 2\alpha \frac{d^2 \mathbf{V}}{dt^2} + \beta \frac{d\mathbf{V}}{dt} = 0, \quad \alpha = \frac{e^2}{2k}, \quad \beta = \frac{3m}{2k}. \quad (77)$$

This is third order, as promised, and being linear it can be solved outright. Setting $\mathbf{W} = d\mathbf{V}/dt$ leaves $\ddot{\mathbf{W}} - 2\alpha \dot{\mathbf{W}} + \beta \mathbf{W} = 0$, with roots

$$\lambda_{\pm} = \alpha \pm \sqrt{\alpha^2 - \beta}, \quad (78)$$

so that

$$\mathbf{V} = \mathbf{A} + e^{\alpha t} \left\{ \mathbf{B} e^{+\sqrt{\alpha^2 - \beta} t} + \mathbf{C} e^{-\sqrt{\alpha^2 - \beta} t} \right\}. \quad (79)$$

Everything now depends on the signs of α and of $\alpha^2 - \beta$, and by Eq. (76) both are controlled by the single number σ :

$$\alpha = \frac{m}{2\sigma e^2}, \quad \alpha^2 - \beta = \frac{m^2}{4\sigma^2 e^4} (1 - 6\sigma). \quad (80)$$

Four regimes follow, and they exhaust the possibilities.

1. $\sigma < 0$. Then $\alpha < 0$ while $1 - 6\sigma > 1$, so $\sqrt{\alpha^2 - \beta} > |\alpha|$ and the two roots straddle zero, $\lambda_+ > 0 > \lambda_-$. If $\mathbf{B} \neq 0$ the motion is self-accelerating. If $\mathbf{B} = 0$ and $\mathbf{C} \neq 0$ it is *self-retarding*: any initial acceleration dies away and the velocity tends to a finite constant. If both vanish the motion is uniform. This is the only regime in which a physically acceptable non-trivial free motion exists.
2. $0 < \sigma < \frac{1}{6}$. Now $\alpha > 0$ and $0 < \sqrt{\alpha^2 - \beta} < \alpha$, so both roots are positive. Any non-zero $\mathbf{B}$ or $\mathbf{C}$ runs away, and the only physical solution is uniform motion.
3. $\sigma = \frac{1}{6}$. The roots coincide and $\mathbf{V} = \mathbf{A} + e^{\alpha t}(\mathbf{B}t + \mathbf{C})$, with $\alpha > 0$; again both constants must vanish.
4. $\sigma > \frac{1}{6}$. Here $\alpha^2 < \beta$, the roots are complex, and

$$\mathbf{V} = \mathbf{A} + e^{\alpha t} \left\{ \mathbf{B} \cos[\sqrt{\beta - \alpha^2} t] + \mathbf{C} \sin[\sqrt{\beta - \alpha^2} t] \right\}. \quad (81)$$

Since $\alpha > 0$ this is an oscillation whose amplitude grows without bound. Eliezer's comment is that such a motion is quite foreign to classical theory and cannot be compared with the wave packets of the quantum theory either.

So the free electron already settles most of the question, analytically and in 1947. Only $\sigma < 0$ admits a non-trivial acceptable motion, and even there it requires $\mathbf{B} = 0$, which is a condition on initial data of exactly the kind Sec. 3.2 objected to in Eq. (LD). Raising the order has not removed the need to impose something by hand; it has added a second constant to impose it on.

5.3. The electron struck by a pulse, solved

The impulse problem of Sec. 3.2 can be done in the same closed form. With a pulse of electric field $\epsilon \delta(t)$,

$$\frac{d^3 v}{dt^3} - 2\alpha \frac{d^2 v}{dt^2} + \beta \frac{dv}{dt} = \gamma \delta(t), \quad \gamma = \frac{3e\epsilon}{2k}, \quad (82)$$

and for $\sigma < 0$, writing $s = \sqrt{\alpha^2 - \beta}$, the solution that stays finite at both ends is

$$v = \begin{cases} \frac{\gamma}{2\beta} \left(1 - \frac{\alpha}{s}\right) e^{(\alpha+s)t}, & t < 0, \\ \frac{\gamma}{\beta} \left[1 - \frac{1}{2} \left(1 + \frac{\alpha}{s}\right) e^{(\alpha-s)t}\right], & t > 0. \end{cases} \quad (83)$$

The two branches agree at $t = 0$, and the exponents have the signs found in Sec. 5.2, so $v \rightarrow 0$ as $t \rightarrow -\infty$ and $v \rightarrow \gamma/\beta$ as $t \rightarrow +\infty$.

That limit is worth evaluating. From Eq. (82),

$$\frac{\gamma}{\beta} = \frac{3e\epsilon}{2k} \cdot \frac{2k}{3m} = \frac{e\epsilon}{m}, \quad (84)$$

which is exactly the velocity a free Newtonian particle of mass m acquires from the same impulse. The equation delivers the right answer at late times, and k has dropped out of it entirely.

What k controls is the approach. The electron is already moving before the pulse arrives, on the timescale $(\alpha + s)^{-1}$, and it is still relaxing afterwards, on the timescale $|\alpha - s|^{-1}$. Eliezer's own remark is that this is *more* symmetrical than the corresponding Lorentz–Dirac solution, in which the electron accelerates until $t = 0$ and then moves uniformly. Set against Fig. 5, this is the same trade in a different currency: the runaway has been removed by a condition imposed at both ends, and the price is again that the effect precedes the cause.

5.4. An observable that was never used

Because Eq. (83) is explicit, so is the radiation it implies. The rate of emission is $\frac{2}{3}e^2(dv/dt)^2$, and Fourier resolving dv/dt gives the energy radiated per unit frequency,

$$E_\nu = \frac{6e^4\epsilon^2}{9(m - 4\pi^2 k\nu^2)^2 + 16\pi^2 e^4\nu^2}, \quad (85)$$

against the corresponding Lorentz–Dirac result, which is Eq. (85) with k set to zero. At low frequency both reduce to $\frac{2}{3}e^4\epsilon^2/m^2$, the Thomson value. At high frequency they part company: Eq. (85) falls as $\frac{2}{3}e^4\epsilon^2 k^{-2}(2\pi\nu)^{-4}$, while the Lorentz–Dirac spectrum falls only as ν^{-2} .

This deserves to be flagged for what it is. It is a difference in a measurable quantity — the cross-section for scattering light off a free electron, at frequencies high enough to matter — between the Lorentz–Dirac equation and its higher-order modification, available in closed form since 1947. Every criterion in Sec. 1.5 concerns the behaviour of solutions; none of them looks at what the particle radiates. The 2006 text, which scanned parameter space numerically for signs of unphysical motion, does not mention Eq. (85), and neither did anything in Secs. 1 to 4.

One further feature is visible in Eq. (85) and is not in C. J. Eliezer (1947). For $k > 0$ the first term of the denominator vanishes at $\nu_0 = (2\pi)^{-1}\sqrt{m/k}$, so the spectrum has a resonance there, with no counterpart in the Lorentz–Dirac case. It is a sharp signature, and it belongs to the sign of k whose free-particle solutions Sec. 5.2 has already excluded.

5.5. The general series

Equation (75) is the shortest admissible extension, and it is where C. J. Eliezer (1947) stops: one extra derivative, one extra constant. The step to a general series is not his. It is taken in D. Zhang (2006), which writes the undetermined vector as a full series in the derivatives of the four-velocity,

$$B_\mu = P_0 v_\mu + P_1 \dot{v}_\mu + P_2 \ddot{v}_\mu + \cdots + P_i v_\mu^{(i)} + \cdots, \quad (86)$$

and asking which terms survive. The instrument is the normalization again. Differentiating $v^\mu v_\mu = -1$ repeatedly gives a chain,

$$\begin{aligned} v_\mu \dot{v}^\mu &= 0, \\ v_\mu \ddot{v}^\mu + \dot{v}_\mu \dot{v}^\mu &= 0, \\ v_\mu \ddot{\dot{v}}^\mu + 3\dot{v}_\mu \ddot{v}^\mu &= 0, \\ v_\mu v^{(4)\mu} + 4\dot{v}_\mu \ddot{\dot{v}}^\mu + 3\ddot{v}_\mu \ddot{v}^\mu &= 0, \\ v_\mu v^{(5)\mu} + 5\dot{v}_\mu v^{(4)\mu} + 10\ddot{v}_\mu \ddot{\dot{v}}^\mu &= 0, \end{aligned} \quad (87)$$

whose coefficients are those of Pascal’s triangle, the identities coming from the Leibniz rule; the general term is Appendix A.4 of D. Zhang (2006). Each identity lets an odd-order derivative of v_μ be traded for lower ones, so the odd terms of Eq. (86) are not independent, and what remains is

$$B_\mu = \left\{ k - \frac{3}{2}P_2(\dot{v})^2 - 5P_4\dot{v}_\mu\ddot{\dot{v}}^\mu - \frac{5}{2}P_4(\ddot{v})^2 - \cdots \right\} v_\mu + P_2\ddot{v}_\mu + P_4v_\mu^{(4)} + \cdots + P_{2i}v_\mu^{(2i)}, \quad (88)$$

Eq. (1.3.2) of D. Zhang (2006), with one undetermined constant for each even derivative retained.

The relation to Sec. 5.1 is worth stating precisely, since the two arguments are easily confused. Eliezer’s conditions kill the $\dot{v}_\mu$ term once, at the first step, and stop there; the identity chain kills every odd term at every order, and so extends the same conclusion to the whole series. Truncating Eq. (88) after P_2 returns Eq. (75), and setting every P_{2i} to zero returns Eq. (LD).

What Eq. (76) said about k applies at every order and gets worse. Each retained P_{2i} is a further constant that the theory does not fix and that no measurement has been proposed to fix, and it arrives together with two more orders of derivative.

5.6. The relativistic equation and the 2006 scans

The 2006 text works relativistically and numerically rather than non-relativistically and analytically. Reducing Eq. (E) to one dimension with the rapidity of Sec. 3.1 gives

$$\ddot{q} - B\ddot{q} + A\dot{q} - \frac{1}{2}\dot{q}^3 = 0, \quad A = \frac{3m}{2k}, \quad B = \frac{e^2}{k}, \quad (89)$$

which is Eq. (77) with $A = \beta$, $B = 2\alpha$, plus the relativistic term $-\frac{1}{2}\dot{q}^3$. The parameters therefore carry the constraints already found: A and B have the same sign, since both are divided by k , and their ratio is fixed,

$$\frac{B}{A} = \frac{2e^2}{3m} = \tau_0 \quad (90)$$

in units with $c = 1$. There is one free number, σ , exactly as Eq. (76) says.

The scan reported in D. Zhang (2006) finds that for every sampled (A, B) the electron accelerates itself to c : monotonically for $A, B < 0$, much as under Eq. (LD), and for $A, B > 0$ with $A \gg B$ by swinging between $+c$ and $-c$ at large amplitude and high frequency. Figures 1-3-1 to 1-3-4 there show $v(\tau)$ for $(A, B) = (2, 1), (3, 1), (10, 1)$ and $(100, 1)$.

Those results are not independent of Sec. 5.2. The oscillatory regime of Eq. (81) is $\alpha^2 < \beta$, which in the present variables reads

$$A > \frac{B^2}{4}, \quad (91)$$

and all four sampled pairs satisfy it. The violent swinging between $\pm c$ is the relativistic form of the growing oscillation Eliezer wrote down in closed form in 1947, with the relativistic term supplying the bound at $\pm c$ that the linear equation lacks. The numerical scan rediscovered an analytic result rather than establishing a new one, and reading the two together sharpens both.

Retaining P_4 as well produces a fifth-order equation, Eq. (1.3.11) of D. Zhang (2006),

$$q^{(5)} + [C + 5\dot{q}^2] \ddot{q} - B\ddot{q} + [A + \frac{25}{2}\dot{q}^2] \dot{q} - \frac{3}{2}\dot{q}^5 - \frac{1}{2}C\dot{q}^3 = 0, \quad (92)$$

with $C = P_2/P_4$, whose sign is free where those of A and B are not. The scan of twenty-two triples, from rest with $q^{(4)}(0) = 2$, produces three classes of behaviour.

Behaviour	Sampled (A, B, C)
Oscillation between $\pm c$	$(100, 1, 1), (100, 1, -1), (100, 1, -100), (1000, 1, -1000), (-1, -100, 100)$
Monotonic approach to c , as under LD	$(-100, -1, 1), (-100, -1, 100), (-100, -1, -10), (-1000, -1, -1000)$
Neither: bounded periodic pulsation	$(100, 1, 100), (100, 1, 1000), (1000, 1, 100), (1000, 1, 1000), (10000, 1, 1000), (-1, -100, 1), (-1, -1000, 10)$

Table 3. The fifth-order scan of D. Zhang (2006), grouped by outcome rather than in its original order. The original table lists $(100, 1, 1)$ twice with contradictory verdicts and repeats three further triples; one of each is kept here, and the class boundaries should be redone before any weight is put on them.

The first two classes are the ones already met. The third is not: the electron neither reaches c nor returns to uniform motion, but settles into a periodic pulsation of the speed within a bounded range — in the 2006 text’s phrase, very like an oscillator, except that the oscillator itself keeps moving forward. This is the case criterion 2 of Sec. 1.5 was written to admit, and it is the one thing in this section that the third-order equation, and hence Eliezer’s analysis, does not contain.

5.7. *What the branch was worth*

Against Sec. 1.5: criterion 1 degrades at every step, each retained P_{2i} adding two orders and two more constants that no measurement supplies; criterion 2 fails at third order for every σ except the negative ones, and there only if a constant is set to zero by hand, and is met by part of the fifth-order parameter space; criterion 3 is never tested at all.

There is also a contemporary verdict that the 2006 text does not engage. G. N. Plass (1961), reviewing the whole subject, held that the standard objections to the Lorentz–Dirac equation were largely invalid, that the extra constants of integration are fixed by requiring that the particle not gain more energy over long times than the external forces supply, and that on that prescription a physical solution exists in every case examined — including, by numerical integration, attractive and repulsive Coulomb fields, where he found no evidence of divergent or unstable solutions. On Eliezer specifically he objected that demonstrating the existence of divergent solutions proves nothing, since divergent solutions always exist, and that a non-divergent physical solution exists for each set of initial conditions. Whether or not one accepts that, it is the position the case for modifying Eq. (LD) has to answer, and neither C. J. Eliezer (1947) nor D. Zhang (2006) answers it.

There is also a sense in which nothing in this section is a finished theory, and it is worth naming precisely, because it is not the usual complaint about higher derivatives.

At Eliezer’s order the equation carries one constant, k , that the theory does not determine. He is candid about having wanted it to determine something — to be a place where Planck’s constant might enter a classical theory — and about that not having worked, so that k has to be written as $\sigma e^4/m$ with σ left to experiment. One undetermined number is a tolerable state of affairs; it is what Eq. (70) also amounts to, and numbers left to experiment can be measured.

The general series is a different matter. Every retained P_{2i} in Eq. (88) is another constant, the theory fixes none of them, no measurement has been proposed for any of them, and there is no principle in sight that would say where to truncate. What Eq. (86) defines is therefore not an equation of motion but a family of them indexed by an infinite sequence of unknowns — a parametrization rather than a theory. That is the respect in which the branch is incomplete, and it is incomplete by construction rather than by oversight: the two conditions on A_μ constrain the series without ever closing it.

What the branch was worth, then, is this. It produced one class of solution not available at lower order, and one observable prediction, Eq. (85), that nobody has used. It cost a free constant and two orders per step, it did not remove the need to impose a condition by hand, and it never closed. And the question the 2006 text closed on —

Would nature really go and choose a more complicated B_μ ? And if it did, we truly cannot imagine the source of nature making such a choice.

— has since been answered, though not in a way available in 2006. That answer belongs with the other twenty-years-on material and is not taken up here.

6. TWENTY YEARS ON

This section pays the debts the previous five deliberately left standing. Section 3.6 set up the fork and declined to say which branch survives. Sections 4 and 5 each followed one branch on its own terms, and each closed by saying that the verdict was not available in 2006. It is available now, and so is the answer to the question D. Zhang (2006) closed on, and so is the fate of its Sec. 5, which declared the discriminating experiment infeasible.

6.1. Why the runaway is not a defect of the equation

The runaway of Sec. 3.2 is not a defect of Eq. (LD). It is what appears when every solution of an effective equation is read as a possible motion, and it disappears as soon as the equation is read as the approximation it is. That claim can be made precise using nothing beyond the one-dimensional form already derived, and this subsection does so.

Take Eq. (43) and solve it for the highest derivative rather than the lowest:

$$\tau_0 \ddot{q} = \dot{q} - \frac{f}{m}. \quad (93)$$

The small parameter multiplies the highest derivative. That is the defining signature of a *singular* perturbation, and it carries a standard consequence: the solutions split into a low-dimensional family on which the motion is slow, and transverse directions on which it is fast.

The fast directions come out in two lines. Put $w = \dot{q}$, let w_s be any solution, and perturb, $w = w_s + \delta$; Eq. (93) then gives $\tau_0 \dot{\delta} = \delta$, so

$$\delta = \delta_0 e^{\tau/\tau_0}. \quad (94)$$

This is Eq. (44) over again, and it has changed status. It is not a solution that has to be forbidden; it is the unstable transverse direction of a singularly perturbed system, and nothing physical is found on it for the reason that nothing physical is found on any unstable direction.

The slow family is equally explicit. Ask for the motions in which $\dot{q}$ is fixed by the applied force alone, and Eq. (93) is solved term by term by

$$\dot{q} = \sum_{n=0}^{\infty} \tau_0^n \frac{d^n}{d\tau^n} \left(\frac{f}{m} \right) = \frac{f}{m} + \tau_0 \frac{d}{d\tau} \left(\frac{f}{m} \right) + \tau_0^2 \frac{d^2}{d\tau^2} \left(\frac{f}{m} \right) + \cdots, \quad (95)$$

as substituting back confirms at once. This is the slow manifold, and it settles the question. It carries no free constant: given the force, the motion is determined by position and velocity alone, which is the Newtonian count. Equation (43) has state $(x, q, \dot{q})$, one dimension more, and that one extra dimension is Eq. (94). One extra dimension, one unstable direction, one datum that Sec. 3.2 had to supply by hand — three descriptions of the same thing.

Two complaints entered earlier can now be withdrawn in their own terms. Section 3.2 objected that $C_1 = 0$ is imposed without warrant; in the present language it says that the data lie on Eq. (95), which is an invariant set the equation itself determines and not a convention. Section 3.3's condition at $\tau = \pm\infty$ was a way of locating that same set without having the concept of it, and the pre-acceleration it introduced is the cost of the indirection rather than a property of the motion.

None of the above is an invention of these notes. The reading of Eq. (LD) as a singularly perturbed system, and of its physical solutions as a manifold rather than a selected subset, is due to H. Spohn (2000) and is set out at length in H. Spohn (2004). What is proved there, for the Abraham–Lorentz model, is that the critical manifold exists, that it is normally hyperbolic, that no solution on it runs away, and that to first order in τ_0 the motion on it obeys

$$\dot{q} = \frac{f}{m} + \tau_0 \frac{d}{d\tau} \left(\frac{f}{m} \right), \quad (96)$$

the truncation of Eq. (95) after the first correction. That equation has a name and a section of its own, and the point to carry into it is that it was not proposed as a replacement for Eq. (LD). It is Eq. (LD)'s own slow manifold, written down.

Where this stands now. The reading above is the standard one and is not in dispute; the disagreements recorded in Secs. 1 to 5, over which modification of Eq. (LD) to adopt, are closed, and they are closed because the premise they shared — that the equation needed modifying — was abandoned. What remains active is narrower and more technical. The series in Eq. (95) is asymptotic rather than convergent, and its structure and resummation have continued to be studied, as has the variational formulation of the equation and its extension to post-Newtonian settings.

6.2. The Landau–Lifshitz equation, and why it has no free parameter

Section 6.1 produced Eq. (96) as the first truncation of a slow manifold. Written covariantly it is

$$ma^\mu = F_{\text{ext}}^\mu + \tau_0 (\delta^\mu{}_\nu + c^{-2}u^\mu u_\nu) \frac{dF_{\text{ext}}^\nu}{d\tau}, \quad (97)$$

the Landau–Lifshitz equation, given in §76 of L. D. Landau & E. M. Lifshitz (1975), with $dF/d\tau$ evaluated on the zeroth-order motion. Its provenance is worth being exact about, because three separate results are doing the work. The equation itself is Landau and Lifshitz’s. That its solutions are the physical ones — that the reduction is not merely a convenient truncation — is H. Spohn (2000). And that it follows from a controlled limit rather than from a choice is S. E. Gralla et al. (2009), discussed below. It is not boxed, and the omission is the point of Sec. 6.1: (AL), (LD) and (E) are candidate laws of motion, and Eq. (97) is not a fourth candidate but a consequence of the second.

Carrying out the differentiation makes the content visible. With $F_{\text{ext}}^\mu = (e/c)F^{\mu\nu}u_\nu$ and $\dot{u}_\nu = F_\nu/m$ to the order kept,

$$\frac{dF_{\text{ext}}^\mu}{d\tau} = \frac{e}{c} \partial_\alpha F^{\mu\nu} u^\alpha u_\nu + \frac{e^2}{mc^2} F^{\mu\nu} F_{\nu\lambda} u^\lambda, \quad (98)$$

so the correction consists of one term in the gradient of the external field and two quadratic in the field itself, the second of which is the radiated four-momentum. No derivative of the acceleration survives anywhere: $\dot{a}^\mu$ has been traded for $\dot{F}_{\text{ext}}^\mu$, which is a property of the applied field and is known before the motion is solved for.

That trade is what buys the good behaviour. Equation (97) is second order, so it is solved from (x, v) like any equation of mechanics; there is no runaway, since Eq. (94) was the transverse direction that has been projected out; and there is no pre-acceleration, since nothing is imposed at $\tau \rightarrow \pm\infty$. All three criteria of Sec. 1.5 are met at once, which Sec. 3.6 showed Eq. (LD) cannot do under any choice of boundary condition. The price is stated plainly by Sec. 6.1: Eq. (97) does not claim to be exact, and holds to first order in τ_0 .

Now the count, which is the subject of this subsection.

Equation	Free parameters	Data beyond (x, v)
Lorentz–Dirac, Eq. (LD)	0	1 per particle
Eliezer, Eq. (E)	1 (σ)	2
The series, Eq. (86)	∞	grows with order
The λ – κ family, Eq. (70)	1 (ζ)	1
Landau–Lifshitz, Eq. (97)	0	0

The coefficient in Eq. (97) is $\tau_0 = 2e^2/3mc^3$, fixed by the charge and the mass and by nothing else, and the same is true of every further term of Eq. (95). This is not the usual situation in an effective theory, where each order brings a coefficient to be measured. It happens here because a classical point charge has nothing to renormalize beyond its mass: once that is absorbed, as in Eq. (32), the expansion is determined.

The tower of Sec. 5.5 therefore collapses, and it is worth being exact about how. Different choices of B^μ correspond to adding to an effective action terms proportional to the equation of motion, and such terms are removed by a redefinition of the field variables, which changes no physical prediction. So the constants P_{2i} of Eq. (88) are not undetermined quantities awaiting measurement. They are a choice of coordinates, and there is nothing there to measure. Section 5.7 closed on the question of whether nature would choose a more complicated B_μ ; the answer is that it does not, because every choice describes the same motion.

Two things make that a result rather than a convention. The first is S. E. Gralla et al. (2009), who obtain the self-force from a controlled point-particle limit — a body of finite size scaled down along a one-parameter family of solutions — with no tube, no cap integrals and no undetermined four-vector. Both reservations of Sec. 2.5 are bypassed rather than answered, since the construction that produced them is never used. The second is Eq. (95) itself, which was constructed without any freedom to choose.

One qualification. All of this is for a structureless point charge. Give the particle structure and genuine parameters appear — a polarizability, a form factor — but they describe the structure, and Sec. 1.3.1 already showed where they live: the terms of Eq. (3) with $n \geq 3$, which the point limit destroys. They are not a radiation-reaction coefficient under another name.

A count of zero is a strong statement, and it invites the obvious question: if the equation has nothing to measure, what is left for an experiment to do? Section 6.3 answers it, and the answer is not that nothing is left.

Finally, a remark that belongs in these notes rather than in the literature. The author's own later drafts, 2010paper.pdf and 2011paper.pdf in this repository, both take Eq. (97) as their starting point and cite Poisson and Rohrlich for it. The move away from the programme of Secs. 4 and 5 was therefore made in 2010, four years after the text those sections reconstruct, and made without comment.

6.3. Which equation the field uses, and why the rest cannot be told apart

Five sections have put nine equations on the table, and two of the nine are families rather than single equations, one of them infinite. Before asking where the classical description stops (Sec. 6.4) and what has been measured (Sec. 6.5), it is worth collecting them in one place and saying what became of each. The answer is short, and it is not the answer Secs. 4 and 5 were working towards.

6.3.1. The roster

Table 4 is every equation these notes have written down.

The list has a shape worth noticing before any verdict is passed on it. Every entry that is third order or higher carries either a free parameter or extra initial data or both; both second-order entries carry neither. That is not a coincidence, and Sec. 6.1 explained it: an extra derivative is an extra solution, and an extra solution has to be selected by something.

Table 4. The candidates, in the order they were introduced.

Equation	Order	Parameters	Status in 2026
Lorentz force	2	0	exact when radiation is negligible
Abraham–Lorentz, Eq. (AL)	3	0	teaching, non-relativistic
Lorentz–Dirac, Eq. (LD)	3	0	the reference law; never integrated
Eliezer, Eq. (E)	3	1 (σ)	abandoned
The B^μ series, Eq. (86)	≥ 3	∞	field redefinitions; no content
λ – κ family, Eq. (70)	3	1 (ζ)	excluded as a theory; see Sec. 6.3.4
Landau–Lifshitz, Eq. (97)	2	0	what everyone uses
Eliezer–Ford–O’Connell, Eq. (89)	2	0	minority alternative
LL with $g(\chi)$, Eq. (104)	2	0	the quantum-corrected mean

NOTE—“Order” is the highest derivative of position. “Parameters” counts numbers not fixed by e and m . The kinetic descriptions of Sec. 6.4 are omitted because they are not equations of motion.

6.3.2. Each candidate was removed by a different argument, and none of them by experiment

Four distinct arguments did the work, and it is worth keeping them separate because they are not interchangeable.

Field redefinition. The B^μ tower is not a set of undetermined constants but a relabelling of what is called the particle’s position at order τ_0 . Every choice describes the same motion, so there is nothing there to measure. This is Sec. 6.2, and it disposes of the infinite family at one stroke.

Energy conservation. The λ – κ family varies the size of the reaction term. That size is not available to be varied: the Larmor formula follows from Maxwell’s equations alone, with no equation of motion anywhere in its derivation, and requiring that the energy the particle loses equal the energy the field carries away fixes the coefficient. Eliezer made this objection in 1947 and it has not been answered since.

Spurious degrees of freedom. The higher-derivative branch of Sec. 5 adds a solution at every order. The parent theory — a body of finite size coupled to its own field — has position and velocity and nothing else, so an equation requiring $\ddot{q}$ as initial data is asking for information that does not exist. This is why the runaways of Sec. 3.2 are not a defect to be repaired but a signal that too many solutions are being carried, and why the reduction of Sec. 6.1 is the cure.

Derivation. What replaced all of it is not an argument against the alternatives at all. S. E. Gralla et al. (2009) obtain the self-force as a controlled limit of a finite body, and what comes out is the form of Eq. (97). Once the equation is derived rather than postulated there is no list to choose from.

None of these is an experimental result. Nothing in the list was excluded by data, and Sec. 6.5 will show why: no experiment has ever been able to tell these equations apart.

6.3.3. Everyone integrates Landau–Lifshitz, for three reasons

The practical answer to “which equation” is a single one. Equation (97) is what is in the particle-in-cell codes, in the strong-field laser modelling, in the pulsar-magnetosphere work, and it is the classical limit against which every quantum treatment is checked. Three separate things put it there.

It is the only one that can be integrated. Equation (LD) needs an initial acceleration, and almost every choice of it runs away; a numerical integration will find the runaway whether or not the analyst wants it. Equation (97) is solved from (x, v) like any equation of mechanics. This is a mundane reason and it is probably the operative one.

Inside the classical domain nothing else is distinguishable. The candidates differ at second order in the expansion parameter $\epsilon = \tau_0 \gamma \omega_B$, and Fig. 7 fixes how large that can get: the line $\epsilon = 1$ lies a factor $3/2\alpha$ above the line $\chi = 1$, so

$$\epsilon|_{\chi=1} = \frac{2\alpha}{3} = 4.9 \times 10^{-3}, \quad \epsilon^2|_{\chi=1} = 2.4 \times 10^{-5}, \quad (99)$$

and at $\chi = 0.01$, where the quantum suppression of the radiated power is still only 5%, $\epsilon^2 \approx 2 \times 10^{-9}$. So in the whole region where the classical description is even approximately valid, the trajectories predicted by Lorentz–Dirac, Landau–Lifshitz and Eliezer–Ford–O’Connell agree to better than one part in 10^5 , and where it is actually accurate they agree to one part in 10^9 . *The choice among them is not an experimental question and never will be.* The disagreement between Landau–Lifshitz and Ford–O’Connell is real, and it is located inside Eq. (99).

It is derived, not selected. This is Sec. 6.3.2, and it is the reason the first two do not feel like a compromise.

6.3.4. The form cannot be tested, so the coefficient is what is left

Equation (99) has a consequence these notes have been building towards without stating it. If the differences among the candidate *forms* are 10^{-5} at the very best and 10^{-9} in practice, then eighty years of argument about which equation is correct has been an argument about something no measurement can reach. Everything in Secs. 1 to 5, and everything in Table 4, lives inside that gap.

What does not live inside it is the coefficient. Write

$$ma^\alpha = F_{\text{ext}}^\alpha + \zeta \tau_0 m (\dot{a}^\alpha - a^\beta a_\beta u^\alpha), \quad \zeta = 1 \text{ predicted}, \quad (100)$$

which is Eq. (70) with $\zeta = -2\kappa$, reduced in order before use. Then ζ multiplies the radiated power directly: double it and the energy loss doubles. It is the one quantity in this problem to which an experiment is linearly sensitive.

The distinction that matters is what ζ is for. It is not a free parameter of any theory on the roster — Lorentz–Dirac, Eliezer, Landau–Lifshitz and Ford–O’Connell all predict $\zeta = 1$, since all of them inherit the coefficient from Larmor. Nor is it a revival of the λ – κ family, which offered a different value as an *alternative* and was excluded for it in Sec. 6.3.2. It is a null-test parameter, introduced for the same reason γ and β are introduced in the parameterised post-Newtonian formalism: general relativity predicts $\gamma = 1$ with no freedom whatever, and one writes γ anyway, in order to say how well the prediction is known.

Said that way the gap is visible. The parameters that are free — the P_{2i} , the order of the equation — change no observable, which is why they can be infinite in number. The parameter that changes

an observable is not free. These are not two statements but one, and the practical upshot is that the only number in this subject that data can constrain is a number nobody has thought to constrain, because theory fixed it and everyone believed the theory.

The state of that constraint is the subject of Sec. 6.5, and the short version is a split: the storage rings of Sec. 6.5.1 measure ζ and nothing else, at $\chi \sim 10^{-6}$, and no one has ever converted seventy years of that data into a bound; the strong-field experiments of Secs. 6.5.4 to 6.5.7 measure $g(\chi)$ while *assuming* $\zeta = 1$, so the joint constraint on the pair has never been examined at all.

6.4. When quantum electrodynamics takes over

Everything so far has been classical, and the question of where that stops has a sharp answer with an uncomfortable consequence: *the classical theory never reaches the field at which it would break down on its own terms*. Quantum mechanics takes over first, and by a fixed factor.

The parameter that decides is

$$\chi = \frac{\gamma B_{\perp}}{B_S}, \quad B_S = \frac{m^2 c^3}{e \hbar} = 4.41 \times 10^{13} \text{ G}, \quad (101)$$

$B_{\perp}$ being the field component transverse to the motion and B_S the Schwinger field. What χ measures is the fraction of the particle's own energy carried off by a single emitted photon. For $\chi \ll 1$ the electron loses its energy in many small pieces, which is what a continuous radiated power describes; for $\chi \gtrsim 1$ a single photon takes a sizeable part of it, and a description in which energy leaks out smoothly is not an approximation to that but a different thing altogether.

Three thresholds are worth separating, since they are usually run together and they fail differently. All three run the same way: the classical description requires the quantity to be *small*, and it is exceeded by going to stronger field or to higher energy, never the reverse. There is no lower boundary to the classical region.

Parameter	Classical requires	What fails once it is exceeded
$\chi = \gamma B_{\perp}/B_S$	$\chi \ll 1$	photon recoil is not small, and the classical emission formula is wrong at leading order, not merely corrected
$b = B/B_S$	$B \ll 4.41 \times 10^{13} \text{ G}$	Landau quantization freezes the transverse motion; there is no gyration left for an equation of motion to describe
$\tau_0 \gamma \omega_B$	$\gamma B \ll 9.07 \times 10^{15} \text{ G}$	classical radiation reaction ceases to be perturbative, and τ_0 merges with the dynamical timescale

Figure 7 puts the three boundaries in the (B, γ) plane. The classical region is the shaded wedge at the bottom left, and one leaves it by going up or to the right.

The third row of the table is the one Secs. 1 to 5 were about. It is also the last of the three to arrive, and the ordering is not an accident of numbers:

$$\frac{B_{\text{cl}}}{B_S} = \frac{m^2 c^4 / e^3}{m^2 c^3 / e \hbar} = \frac{\hbar c}{e^2} = \frac{1}{\alpha} \approx 137. \quad (102)$$

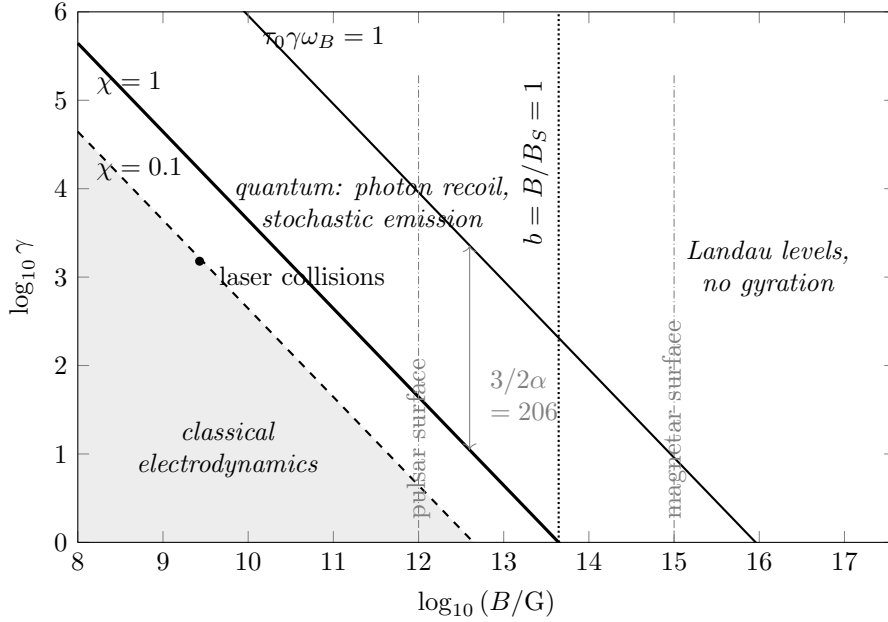


Figure 7. Where classical electrodynamics applies. Shaded is the region in which it does: $\chi \lesssim 0.1$ and $B \lesssim B_S$. Crossing *upward or to the right* across any boundary leaves it, and the three boundaries fail in different ways. Above $\chi = 1$ a single photon carries a sizeable part of the particle’s energy, so the emission is quantum and stochastic. To the right of $B = B_S$ the transverse motion is Landau quantized and there is no gyration for an equation of motion to describe. The line $\tau_0 \gamma \omega_B = 1$ is where classical radiation reaction would itself become non-perturbative, which is the regime Secs. 1 to 5 are about — and it lies a factor $3/2\alpha = 206$ above the line where the classical description of the emission has already failed, so it is never reached. Laser–electron collisions sit at $\chi \approx 0.09$ for $a_0 = 10$ at 800 nm against a $\gamma = 1500$ beam, the relevant field being the $2\gamma E$ seen head-on, just inside the classical region and close enough to the boundary to test it. Pulsar and magnetar surface fields are marked; at a magnetar surface even a mildly relativistic electron is already past both quantum boundaries.

The same statement can be made without reference to fields at all. The classical electron radius and the reduced Compton wavelength stand in the ratio

$$\frac{r_0}{\lambda_C} = \frac{e^2/mc^2}{\hbar/mc} = \alpha, \quad (103)$$

so the length at which classical electrodynamics runs into itself lies inside the length at which quantum mechanics has already taken over, and it lies inside by exactly α . Equation (102) is Eq. (103) in the language of fields.

The consequence is worth stating flatly, because it settles the motivation of everything before this section. The runaway of Sec. 3.2, the pre-acceleration of Sec. 3.3, the merging of τ_0 with the physical timescale — these become $O(1)$ effects only at B_{cl} , and B_{cl} is 137 times past the point where the classical description of the emission has already failed. *The classical theory does not get to break down.* It is superseded well before its own pathologies become measurable, which is why no experiment has ever seen a runaway and why none ever will.

Where classical electrodynamics is still the right tool is correspondingly specific: $\chi \lesssim 0.1$. Three places qualify. Laser–electron collision experiments work deliberately at $\chi \sim 0.1$ to 1, which is the

transition region and the only place the strength of the radiation reaction can be measured against a classical prediction. The outer regions of pulsar magnetospheres, far from the surface, have B low enough for classical treatment, and that is where the Aristotelian electrodynamics of Sec. 4.4 lives. And the classical result remains the $\chi \rightarrow 0$ limit that any strong-field QED calculation has to reproduce, which is how such calculations are checked.

None of this says that classical radiation reaction is wrong. Section 6.2 stands and Eq. (97) is exact in its domain; what the domain excludes is the regime in which the classical theory's own difficulties would have become visible. Nor does it say the strong-field problem is uninteresting. It says the problem there is a different one, and it is worth setting down what that problem is, because it is where the subject went.

Past $\chi \sim 1$ the classical emission formula is replaced rather than corrected. The radiated power becomes

$$P = g(\chi) P_{\text{cl}}, \quad g(\chi) \simeq 0.56 \chi^{-4/3} \quad (\chi \gg 1), \quad (104)$$

with g the quantum suppression factor, and at $\chi = 1$ the true power is already about half the classical value. That much could be absorbed into an equation of motion by rescaling a coefficient, and it is tempting to read it as the measurement of ζ that Sec. 4.5 asked for. It is not, and the reason is the second change, which no rescaling reproduces.

Emission is not continuous. The electron radiates one photon at a time, and once a single photon carries an appreciable fraction of the energy, two electrons prepared identically do not follow the same trajectory: the energy loss is a random variable and the final state is a distribution. This is straggling, and it is not a correction to a deterministic law but a statement that the law is the wrong kind of object. *No equation of motion, of any order, with any coefficient, produces a spread from identical initial data.* Sections 4 and 5 argued about which ordinary differential equation to write down; past $\chi \sim 1$ the answer is none of them.

What is used instead is a kinetic description: photon emission sampled event by event from the strong-field QED rates, usually under the locally-constant-field approximation, with the particle pushed classically between emissions; or a Fokker–Planck treatment when the second moment suffices; or Eq. (104) inserted into Eq. (97) when only the mean matters and the spread does not. The locally-constant-field approximation is itself the subject of current work, since it is known to fail for the low-energy part of the emitted spectrum.

6.5. What has actually been measured

Sections 4 and 5 argued on the only grounds available to them, whether the solutions of a proposed equation look physical. That is not the only ground available, and it is worth setting out what has been measured, because the experimental record is both older and broader than the recent laser work suggests, and most of it sits in the classical region of Fig. 7 rather than outside it.

Each subsection below is one experiment or one class of them, in order of increasing χ , and each ends with the figure in the primary paper that carries the result, so that the claim can be checked rather than taken on trust. Table 5 collects the same information.

6.5.1. Storage rings have been measuring the coefficient for seventy years

Every electron storage ring is a radiation-reaction experiment that runs continuously, and the quantity it measures is the coefficient of Eq. (97) and nothing else. On a circular orbit of radius ρ

the Larmor formula integrates over one turn to an energy loss

$$U_0 = \frac{4\pi}{3} \frac{e^2 \gamma^4}{\rho}, \quad (105)$$

which is $\frac{2}{3}e^2\dot{v}^2/c^3$ carried around the ring, so the number in front of it is the same τ_0 that Secs. 1 to 5 have been arguing about. That loss is what damps the betatron and synchrotron oscillations, with damping times

$$\tau_i = \frac{2E_0 T_0}{J_i U_0}, \quad J_x + J_y + J_s = 4, \quad (106)$$

T_0 being the revolution period and J_i the partition numbers. The sum rule is Robinson's theorem (K. W. Robinson 1958): the damping can be moved between the three degrees of freedom by changing the lattice, but the total is fixed by the radiated power alone. Both parts of Eq. (106) are measured in every machine that is commissioned. For a 3 GeV ring with $\rho \sim 10$ m the loss is ~ 0.7 MeV per turn and the damping time a few milliseconds, some thousands of turns, which is short enough that a ring that did not damp would not store beam at all.

The regime is deeply classical: $\gamma \approx 6 \times 10^3$ in a $B = 10^4$ G bending field gives $\chi \approx 1.3 \times 10^{-6}$, at the far lower left of Fig. 7. That is exactly why the result is worth stating, because it settles a question the theoretical literature sometimes leaves open. *Radiation reaction is not a conjectural effect*, and a measured damping time is a measurement of τ_0 on a real beam. The disputes of Secs. 3 to 5 are disputes about what to do with a term whose size was settled experimentally before any of them began.

Where to look. There is no single canonical paper or figure here, which is itself the point: it is accelerator engineering, not a result. The standard derivation of Eqs. (105) and (106) is M. Sands (1970); the theorem is K. W. Robinson (1958).

6.5.2. The same machines have measured the granularity of the emission since the 1970s

Those rings also measure the quantity Sec. 6.4 called the signature of the quantum regime. Emission is by photons, so the energy loss fluctuates, and the equilibrium energy spread is set by the balance of that fluctuation against the damping of Eq. (106):

$$\left(\frac{\sigma_E}{E}\right)^2 = \frac{C_q \gamma^2}{J_s \rho}, \quad C_q = \frac{55}{32\sqrt{3}} \frac{\hbar}{mc} = 3.83 \times 10^{-13} \text{ m}. \quad (107)$$

The constant is proportional to $\hbar$ and to nothing else, so a beam with a measured energy spread is a measurement of the granularity: were the emission continuous, σ_E would be zero. The same shot noise sets the horizontal emittance, and the spin polarizes radiatively by the Sokolov–Ternov effect (A. A. Sokolov & I. M. Ternov 1964) to an asymptotic $8/5\sqrt{3} = 92.4\%$, first seen at ACO and VEPP-2 around 1971.

The consequence for Sec. 6.4 is that stochastic emission is not a strong-field novelty. It has been observed for half a century at $\chi \sim 10^{-6}$, where it is a small diffusion superposed on a deterministic drift and where Eq. (107) describes it completely. What changes at $\chi \sim 1$ is not that the emission becomes stochastic — it always was — but that the diffusion stops being small compared with the drift, so that the spread can no longer be carried as a correction to a deterministic trajectory.

Where to look. M. Sands (1970) again for Eq. (107); the figure worth reproducing is a polarization build-up curve from any of the e^+e^- rings, which shows the exponential approach to 92.4% on the Sokolov–Ternov time scale.

6.5.3. *A single trapped electron shows the effect is not collective, and that it depends on the boundary*

A beam is many particles, and one could ask whether the damping is a collective effect. It is not. The cyclotron motion of one electron in a Penning trap damps radiatively at

$$\gamma_{\text{damp}} = \tau_0 \omega_c^2, \quad \omega_c = \frac{eB}{mc}, \quad (108)$$

which is the Abraham–Lorentz damping rate of Eq. (AL) evaluated on a circular orbit and nothing more; it is the rate for the amplitude, so the cyclotron energy decays at $2\tau_0\omega_c^2$. At $B = 5$ T, $\omega_c/2\pi = 140$ GHz and Eq. (108) gives an energy lifetime of about 0.1 s, long enough to watch directly. It is measured on one particle, which is as clean a test of the coefficient as exists.

The interesting part is that the measured rate is not always the free-space one. G. Gabrielse & H. Dehmelt (1985) found the decay inhibited by more than an order of magnitude when the cyclotron frequency was tuned away from a mode of the microwave cavity formed by the trap electrodes. This is the first observation of inhibited spontaneous emission, and it makes a point that matters for these notes: the self-force is not a property of the particle. It is a property of the particle together with the field it radiates into, and changing the boundary conditions on that field changes it. Nothing in Secs. 2 to 5 could have anticipated this, because all of them work in unbounded Minkowski space.

Where to look. G. Gabrielse & H. Dehmelt (1985), the figure giving the measured cyclotron decay rate as the trap is tuned through the cavity resonances; the suppression is read off directly against the free-space value of Eq. (108).

6.5.4. *E-144 did in 1997 the experiment the thesis said could not be done*

The first strong-field experiment predates D. Zhang (2006) by nearly a decade. At the SLAC Final Focus Test Beam a 46.6 GeV electron beam was collided with terawatt pulses of 527 nm light at $\sim 10^{18}$ W/cm², giving $\eta = 0.36$ in the notation of Eq. (110) — their η is a_0 — and $\Upsilon = 0.84\eta = 0.3$, their Υ being χ . Two results came out of it. C. Bula et al. (1996) observed nonlinear Compton scattering, an electron absorbing several laser photons in one event. D. L. Burke et al. (1997) then observed electron–positron pairs from the collision of the backscattered GeV photon with several more laser photons, 106 ± 14 positrons above background, and this was the first observation of inelastic light-by-light scattering with real photons only. The yield scaled as

$$R_{e^+} \propto \eta^{2n}, \quad n = 5.1 \pm 0.2 \text{ (stat)}_{-0.8}^{+0.5} \text{ (syst)}, \quad (109)$$

the fifth power of the intensity, which is the multiphoton threshold made visible: the process needs at least five photons and the rate reports that number back.

This is not a radiation-reaction measurement — the electrons emitted at most a few photons, so nothing recoiled appreciably — but it establishes that $\chi \sim 0.3$ was experimentally accessible in 1997, and it is the direct counterexample to the conclusion of Sec. 5 of D. Zhang (2006) that this regime was out of reach.

Where to look. D. L. Burke et al. (1997), Fig. 4: positron yield per laser shot against η on log axes, with the power-law fit of Eq. (109) and the shaded 95% background limit. Fig. 3 of the same paper is the raw laser-on minus laser-off positron spectrum.

6.5.5. Aligned crystals reach $\chi \sim 1$ with no laser at all

A crystal aligned to a beam presents a channelled particle with the coherent field of an atomic string, of order 10^{11} V/cm, and in the rest frame of a 100 GeV particle that is already comparable with B_S . T. N. Wistisen et al. (2018) sent 178.2 GeV positrons along the $\langle 111 \rangle$ axis of silicon crystals 3.8 and 10.0 mm thick, reaching $\chi \lesssim 1.4$, and measured the emitted photon spectrum. In that regime each positron emits several photons carrying an appreciable fraction of its own energy, so the recoil is not a perturbation and the spectrum is sensitive to how it is treated. They compared four calculations — classical with radiation reaction, semiclassical, quantum with radiation reaction, and quantum without — and only the quantum treatment including recoil reproduced the data.

The platform matters because its systematics have nothing in common with a laser collision. The field is static in the lab, the target is a solid with a known thickness, and the number of interacting particles is not in doubt; there is no beam overlap to estimate, which is the dominant uncertainty in everything in Sec. 6.5.6. An agreement between crystal and laser data is therefore a stronger statement than either alone.

Where to look. T. N. Wistisen et al. (2018), Fig. 3: background-subtracted photon power spectrum for the two thicknesses, panels (a) and (b), with the four model curves overlaid. Fig. 4 of the same paper shows the four models on their own, without the detector response folded in, which is the cleaner figure for seeing how far apart the classical and quantum predictions actually are.

6.5.6. The 2018 laser collisions saw radiation reaction but could not separate the models

The modern line begins with two experiments on the same laser in the same year. The controlled quantities are

$$a_0 = \frac{eE}{mc\omega}, \quad \chi = \frac{2\gamma a_0 \hbar \omega}{mc^2}, \quad (110)$$

the factor 2γ being the boost of the laser field into the electron rest frame in a head-on collision; $a_0 = 10$ at 800 nm against a $\gamma = 1500$ beam gives $\chi \approx 0.09$, the point marked in Fig. 7.

J. M. Cole et al. (2018) collided a laser-wakefield beam above 500 MeV with a pulse at $a_0 > 10$ and measured two things at once: the electron energy after the collision and the critical energy of the γ -ray spectrum, above 30 MeV. Neither alone is conclusive, because a_0 is not known shot to shot, but the *correlation* between them is, since each model of radiation reaction predicts a different curve in the $(\varepsilon_{\text{final}}, \varepsilon_{\text{crit}})$ plane parameterised by a_0 along its length. The data preferred a quantum description. K. Poder et al. (2018) pushed to beams above 2 GeV at 4×10^{20} W/cm², where the electrons see about a quarter of the critical field, and lost up to 30% of their energy; comparing against four models — Lorentz force only, Landau–Lifshitz, semiclassical, and full QED — the classical Landau–Lifshitz curve overshot the loss and the two quantum treatments did not.

Neither experiment separated the quantum treatments from one another, and neither was a high-significance exclusion of the classical one. The reason was the same in both: the dominant uncertainty is the spatial and temporal overlap of two beams a few microns across, which is not a statistical error and does not improve with more shots.

Where to look. J. M. Cole et al. (2018), Fig. 9 of arXiv:1707.06821: $\varepsilon_{\text{crit}}$ against $\varepsilon_{\text{final}}$, four data points, with shaded bands showing what each model would produce for a_0 uniform on $[4, 20]$ — the bands overlap, which is the honest picture of the discrimination available in 2018. K. Poder et al.

(2018), Fig. 4 of arXiv:1709.01861: the measured post-collision electron spectrum against the four models in panels (a)–(d), which is the single most legible model-comparison figure in this literature.

6.5.7. *The 2026 measurement separates the models, and the classical one loses*

The discrimination that 2018 could not make has now been made. E. E. Los et al. (2026) report the first observation of strong-field radiation reaction at above 5σ in which quantum effects matter, using a 609 MeV beam against a pulse of peak $a_0 = 21.4 \pm 1.8$, with a_0 between 7 and 13 inferred for the shots analysed. The method is what the 2018 work lacked: over 600 collisions and 608 null shots, so that the difference between hit and null distributions can be established statistically, followed by a Bayesian inference on the ten highest-yield shots that treats the unknown a_0 as a parameter to be marginalised rather than a systematic to be quoted. The result is a Bayes factor favouring the quantum-continuous and quantum-stochastic models over the classical one, driven by the classical model predicting too much energy loss. The two quantum models are still not separated from each other.

The parameter to note is χ : at most 0.13 at focus and 0.05 to 0.1 for the shots used. That is not deep in the quantum regime; it is the $\chi = 0.1$ line of Fig. 7. The classical description of the emission fails, measurably and at high significance, exactly where that figure says it should, and this is the strongest evidence available that the boundary drawn there is the physical one.

Where to look. E. E. Los et al. (2026), arXiv:2407.12071. Fig. 4: the measured post-collision electron spectrum for the highest-yield shot against the three models. Fig. 5: the Bayes factors, shot by shot and combined, which is the figure that carries the 5σ claim and the one to reproduce if only one is wanted.

6.5.8. *LUXE and E-320 are being built to scan χ through unity*

Everything above either sits well below $\chi = 1$ or, in the crystal case, reaches it in a target that cannot be tuned. Two experiments now under construction are designed to sweep through it. LUXE at DESY will collide the 16.5 GeV European XFEL beam with a laser, and E-320 at SLAC will do the same with the 10 GeV FACET-II beam; both can vary the laser intensity at fixed beam energy, so that χ becomes a knob rather than a shot-to-shot accident. That is what is needed to test $g(\chi)$ of Eq. (104) as a function rather than at a point, and to reach the regime where the locally-constant-field approximation of Sec. 6.4 can be checked against data.

Three things follow from the list. The first is that the existence of radiation reaction, and the accuracy of its classical description where that description applies, are not open questions and have not been for seventy years; the rings of Sec. 6.5.1 settle it, on beams and, in Sec. 6.5.3, on one particle at a time. The second is that the strong-field regime was reached in 1997, not in 2018, and the 2006 assessment of what was feasible was already nine years out of date when it was written. The third is that the classical description has now been excluded, at 5σ and at $\chi \approx 0.1$, which is the first time any of the equations in these notes has been ruled out by data rather than by argument — and what was ruled out was Eq. (97), the survivor of Secs. 1 to 5, in the one place where it was ever going to fail.

6.6. *Open questions, and one that nobody has attempted*

[Separate the settled from the open, and be honest about which is which. Settled: (LD) is effective; reduction of order is its physical content; the B^μ freedom is a field redefinition; the rigorous derivation

Table 5. The experimental record, in order of increasing χ .

Experiment	When	χ	Measured	Fig.
Penning trap	1985	10^{-9}	cyclotron damping, one electron	—
Storage rings	1950s–	10^{-6}	damping times, Robinson rule	—
Quantum excitation	1970s–	10^{-6}	energy spread, polarization	—
E-144	1996–97	0.3	Compton, pair creation	4
Cole 2018	2018	~ 0.1	$\varepsilon_{\text{crit}}$ vs. $\varepsilon_{\text{final}}$	9 ^a
Poder 2018	2018	~ 0.25	electron spectrum	4 ^a
Wistisen 2018	2018	$\lesssim 1.4$	photon spectrum	3
Los 2026	2026	0.05–0.1	Bayes factors, $> 5\sigma$	4, 5 ^a
LUXE, E-320	building	through 1	$g(\chi)$ as a function	—

NOTE—Full references in the text, Secs. 6.5.1 to 6.5.8. ^aFigure numbers of the preprint version, which is the version checked here. The first three rows are standing accelerator practice rather than single experiments and have no canonical figure.

exists; the singular field exerts no force. Open: (i) which description fits the data in the transition region $\chi \sim 0.1$ to 1 — classical, quantum-continuous with a Gaunt factor $g(\chi)$, or stochastic emission; (ii) how tightly the coefficient itself is bounded, since ζ of Eq. (70) is fixed in theory but has never been constrained by data — this is the most concrete of the five; (iii) where Landau–Lifshitz and Ford–O’Connell part company at higher order in τ_0 , and whether that regime is inside the classical domain at all, which it probably is not, making the disagreement moot; (iv) the asymptotic character of the τ_0 expansion and its resummation; (v) on the quantum side, $g(\chi)$ at $\chi \gg 1$ and the failure of the locally-constant-field approximation at low harmonics.]

6.7. What survives is the coefficient, not the equation

[Settle the naming debt here, moved out of Sec. 6.1, since it is a statement about what survived: comparing Eq. (96) with Eq. (89), both start from Eq. (LD), one replaces $\ddot{q}$ by a derivative of the force and lowers the order, the other adds $\ddot{q}$ and raises it, and the two carry the same name — the first being the Eliezer–Ford–O’Connell equation (C. J. Eliezer 1948; G. W. Ford & R. F. O’Connell 1991), free of runaways and a live experimental candidate. The upward branch of Sec. 5 does not survive, and it runs opposite to the cure: each retained order adds an unphysical mode rather than removing the one already there. The sideways branch of Sec. 4 does. Strip the scaffolding and what is left, Eq. (70), is a number that data can constrain, and the transition region of Sec. 6.2 is where it can be constrained. And the Aristotle motion of Sec. 4.4 survives under another name, as the Aristotelian electrodynamics now used in pulsar magnetosphere modelling, where the field fixes the velocity rather than the acceleration. Close on the shape of the whole thing: the pathology these notes tracked for five sections turns out to be an artefact of treating an effective equation as a fundamental one, while the parameter invented to cure it is the only part that can be measured.]

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